

EE 6106

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1 Prediction via Expert Advice

The problem here is to predict a sequence X_t that is being generated by some *all-knowing adversary*. First, let us deal with predicting binary sequences $X_t \in \{0, 1\}$. The general framework is as illustrated below.

- History of the actions is known and history of expert decisions is known.
- Learning agent at time t gets $Y_{i,t}$ which is the output/decision of expert i at time t . Agent gets the output of all the experts at time t .
- Agent takes action A_t based on expert outputs.
- Then the actual “ground truth” is revealed.
- **GOAL:** Over an horizon, we need to minimize some loss function like $\sum_{t=1}^n \mathbb{1}\{X_t \neq A_t\}$

Considering the sequence is generated by an adversary, can we guarantee a bound on the worst-case loss?

Majority Algorithm (MA)

Assumption 1. *There is a perfect expert i.e. always gives the “ground truth”.*

This assumption is strong, but gives a very nice strategy. Keep track of a set of experts S who haven't made a mistake yet. Take A_t to be the majority of the output of the experts in S .

Algorithm 1 Majority Algorithm

Require: Number of expert K , horizon n

- 1: **Initialize:** $S_0 = \{1, 2, \dots, K\}$
 - 2: **for** $t = 1$ to n **do**
 - 3: $A_t = \text{Majority}(Y_{i,t})$
 - 4: Observe X_t
 - 5: $S_{t+1} = S_t \setminus \{i : Y_{i,t} \neq X_t\}$
 - 6: **end for**
-

Claim 1.1. *The loss under the majority expert is at most $\log_2 K$ for K experts.*

Proof. Since we are taking majority vote, the worst case scenario becomes that half of the experts make a mistake. However, in this case, we can remove half of the experts. Thus, the bound will be $\log_2 K$.

In case if only one expert is making a mistake at each time step, then the majority vote will be

correct so the agent never makes a mistake. $\square$

The mistake bound is **independent** ($O(1)$) of the horizon n and purely depends on the number of experts we have.

Weighted Majority Algorithm (WMA)

Now, let us **disregard** Assumption 1 i.e. there is no perfect expert. In this case, how do we ensure we bound the error, given the fact that an adversary with complete knowledge of the algorithm decides the input sequence? The answer is a **softer version** of the majority algorithm. In this case, we give all experts a weight of 1 to begin with, which defined the confidence we have in the ability of said expert. However, as and when an expert makes a mistake, we decay their weight by some factor, indicating a reduction in the confidence of their decision.

Algorithm 2 Weighted Majority Algorithm

Require: Number of expert K , horizon n , learning rate $\beta \in (0, 1)$

```
1: Initialize: Weight vector  $W_0 = \{1, 1, \dots, 1\}$ 
2: for  $t = 1$  to  $n$  do
3:   Sum of weights of experts predicting 1 -  $w_1 = \sum_{i=1}^K W_t[i] * \mathbb{1}[Y_{i,t} = 1]$ 
4:   Sum of weights of experts predicting 0 -  $w_0 = \sum_{i=1}^K W_t[i] * \mathbb{1}[Y_{i,t} = 0]$ 
5:    $A_t = \mathbb{1}\{w_1 \geq w_0\}$ 
6:   Observe  $X_t$ 

7:   for  $i = 1$  to  $K$  do
8:     if  $Y_{i,t} \neq X_t$  then
9:        $W_{t+1}[i] = W_t[i]$ 
10:    else
11:       $W_{t+1}[i] = (1 - \beta)W_t[i]$ 
12:    end if
13:  end for

14: end for
```

Note

1 is a **special case** of 2 where $\beta = 1$.

Also note that, the adversary knows 2 and thus can ensure you make a mistake at each time step. However, that means that there are experts that are making a mistake at each time step. Thus, it is unreasonable for the agent to perform well when the experts are messing up. Thus, we benchmark against the performance of the experts.

Let us now get bounds on the error for Algorithm 2. Define -

$$W_t = \sum_{i=1}^K W_{i,t}$$

as the sum of weights of all experts at the beginning of time step t . The general idea is to bound the sum of weights W_{n+1} after a horizon n .

Lower Bound

Claim 1.2. $W_{n+1} \geq (1 - \beta)^{L_{i,n}}$

Where $L_{i,n}$ is the loss of expert i which is the number of mistakes it makes. So, the RHS is actually the weight of the i^{th} expert after the horizon. So the bound is trivial and very very loose (like you mum hehe). But turns out this is good enough.

Upper Bound

Define that the total weight at time t is a sum of “good” weight (experts who didn’t make a mistake) and “bad” weight (experts who made a mistake).

$$\begin{aligned} W_t &= GW_t + BW_t \\ W_{t+1} &= GW_t + (1 - \beta)BW_t = (1 - \beta)W_t + \beta GW_t \end{aligned}$$

We also know that (because of majority), after a mistake, we get -

$$GW_t \leq \frac{1}{2}W_t$$

Thus, we get -

$$\begin{aligned} W_{t+1} &\leq (1 - \beta)W_t + \frac{\beta}{2}W_t \\ \implies W_{t+1} &\leq \left(1 - \frac{\beta}{2}\right)W_t \\ \implies W_{n+1} &\leq \left(1 - \frac{\beta}{2}\right)^{L_n} \cdot K \end{aligned}$$

The total number of mistakes is L_n and the initial weight $W_0 = K$. Now we have both lower and upper bounds, which gives us -

$$\begin{aligned} \left(1 - \frac{\beta}{2}\right)^{L_n} \cdot K &\geq (1 - \beta)^{L_{i,n}} \\ \ln K + L_n \ln \left(1 - \frac{\beta}{2}\right) &\geq L_{i,n} \ln(1 - \beta) \end{aligned}$$

We know the inequality -

$$1 - \frac{1}{x} \leq \ln x \leq x - 1 \quad \forall x > 0$$

Then, we can write taking $x = 1 - \frac{\beta}{2}$, we get -

$$\begin{aligned} 1 - \frac{1}{\left(1 - \frac{\beta}{2}\right)} &\leq \ln \left(1 - \frac{\beta}{2}\right) \leq 1 - \frac{\beta}{2} - 1 \\ \frac{\frac{\beta}{2}}{\frac{\beta}{2} - 1} &\leq \ln \left(1 - \frac{\beta}{2}\right) \leq -\frac{\beta}{2} \\ \ln K - \frac{\beta L_n}{2 - \beta} &\leq \ln K + L_n \ln \left(1 - \frac{\beta}{2}\right) \leq \ln K - \frac{\beta L_n}{2} \end{aligned}$$

Thus, we can write -

$$\begin{aligned} \ln K - \frac{\beta L_n}{2} &\geq L_{i,n} \ln(1 - \beta) \\ \frac{\beta}{2} L_n &\leq \ln K - L_{i,n} \ln(1 - \beta) \\ L_n &\leq \frac{2}{\beta} \ln K - \frac{2}{\beta} L_{i,n} \ln(1 - \beta) \end{aligned}$$

We know from the inequality again that -

$$\ln(1 - \beta) \geq 1 - \frac{1}{1 - \beta} = \frac{-\beta}{1 - \beta}$$

Thus, we get -

$$L_n \leq \frac{2}{\beta} \ln K - \frac{2}{\beta} L_{i,n} \frac{-\beta}{1 - \beta}$$

This gives us the following bound -

$$L_n \leq \left(\frac{2}{1 - \beta} \right) L_{i,n} + \frac{2}{\beta} \ln K \quad (1)$$

This is true for any expert i . That means, the tightest bound is found when we take i to be the best expert.

$$L_n \leq \left(\frac{2}{1 - \beta} \right) \left[\min_i L_{i,n} \right] + \frac{2}{\beta} \ln K \quad (2)$$

To measure how good our algorithm performs, we define the concept of **regret** of the algorithm i.e. how bad does it perform with respect to the best expert¹.

$$\text{Regret} = R_n = L_n - \left[\min_i L_{i,n} \right]$$

In the case of Algorithm 2, we get -

$$\begin{aligned} R_n^{\text{WMA}} &\leq \left(\frac{2}{1 - \beta} \right) \left[\min_i L_{i,n} \right] + \frac{2}{\beta} \ln K - \left[\min_i L_{i,n} \right] \\ &\leq \left(\frac{1 + \beta}{1 - \beta} \right) \left[\min_i L_{i,n} \right] + \frac{2}{\beta} \ln K \end{aligned}$$

Ideally, we need the regret to be **sub-linear** with respect to the horizon n so that asymptotically, the regret at each time step tends to zero. The regret in the case of Algorithm 2 will be sub-linear **IFF** the loss of the best expert i.e. $\min_i L_{i,n}$ is sub-linear.

Claim 1.3. *For the given formulation of the 0 – 1 loss, it is **impossible** to get a **deterministic** algorithm A that gives sub-linear regret.*

Proof. The proof of this statement is quite intuitive. Suppose we have a deterministic algorithm A that chooses the experts at each time step based on some pre-determined condition. Then, the adversary (which we have assumed to be all-knowing) has can conspire to ensure that A incurs loss at each time step. Consider a simple example - $K = 2$ such that expert 1 always predicts 1 and expert 2 always predicts 0. Then, the best case regret -

$$R_n^{\text{best}} \geq \frac{n}{2} = \Theta(n)$$

□

In other words, to get sub-linear regret, we need either a **softer loss** when compared to 0 – 1 loss or a **stochastic** algorithm.

¹in hindsight since we can find the best expert after the entire process is done.

Exponential WMA

Now, we transition from predicting purely binary sequences to predicting real-valued sequences. Let $X_t, Y_{i,t}, A_t \in [0, 1]$ be real numbers. Define a loss function $\ell(X_t, A_t) \in [0, 1]$ that is the loss incurred by the agent when taking action A_t at time t when the true sequence was X_t . We shall also assume that ℓ is **convex** in the second variable i.e. A_t . Now, we can define a new algorithm that minimizes -

$$L_n = \sum_{t=1}^n \ell(X_t, A_t)$$

Algorithm 3 Exponential WMA

Require: Number of expert K , horizon n , learning rate $\beta \in (0, 1)$

```

1: Initialize: Weight vector  $W_0 = \{1, 1, \dots, 1\}$ 
2: for  $t = 1$  to  $n$  do
3:   Sum of weights of experts predicting 1 -  $w_1 = \sum_{i=1}^K W_t[i] * \mathbb{1}[Y_{i,t} = 1]$ 
4:   Sum of weights of experts predicting 0 -  $w_0 = \sum_{i=1}^K W_t[i] * \mathbb{1}[Y_{i,t} = 0]$ 
5:    $A_t = \frac{\sum_{i=1}^K W_t[i] \cdot Y_{i,t}}{\sum_{i=1}^K W_t[i]}$ 
6:   Observe  $X_t$ 
7:   for  $i = 1$  to  $K$  do
8:      $W_{t+1}[i] = W_t[i] e^{-\beta \cdot \ell(X_t, Y_{i,t})}$ 
9:   end for
10: end for

```

Note here that A_t is no longer a binary value, but is a real number. It is a weighted probability of all experts and at each time step, the weights of all the experts are being updated. We also assume $\beta > 0$.

- In Algorithm 3, the action is a **real number** between $[0, 1]$ instead of a binary number. This saves us from the “all-or-nothing” formulation we had used in the previous algorithms.
- Another thing is that the weights are being updated exponentially. If the loss is low, then the decay in weight is low. However, higher losses incur higher decays. Higher β makes the system more “sensitive” to errors as smaller losses also cause larger decays.

Now, we can perform the regret analysis of Algorithm 3. To do so, we first need to define a few mathematical tools that shall help us from here on out.

Hoeffding's Lemma

Lemma 1.4. For a bounded random variable $X \in [a, b]$, we can upper bound the MGF as follows -

$$\mathbb{E}[e^{SX}] \leq \exp \left\{ S\mathbb{E}[X] + \frac{1}{8}S^2(b-a)^2 \right\}$$

If $X \in [0, 1]$ like most RVs representing probability values, we can simplify the expression as -

$$\mathbb{E}[e^{SX}] \leq \exp \left\{ S\mathbb{E}[X] + \frac{1}{8}S^2 \right\}$$

Proof. We make use of a *symmetrization technique* to prove this expression. Consider X' to be a copy of X such that $X' \in [a, b]$ and $\mathbb{E}[X'] = \mathbb{E}[X]$ but X, X' are independent to each other. Then,

$$\mathbb{E}_X [\exp \{S(X - \mathbb{E}_X[X])\}] = \mathbb{E}_X [\exp \{S(X - \mathbb{E}_{X'}[X'])\}] \leq \mathbb{E}_X [\mathbb{E}_{X'} \exp \{S(X - X')\}]$$

This uses the fact that e^{-x} is convex and uses Jensen's inequality. □

Let us now get the lower and upper bounds for Algorithm 3.

Lower Bound

Consider the term $W_t = \sum_{i=1}^K W_t[i]$ which is the sum of weights of the experts at time t . For the lower bound, we perform a similar analysis as the previous case and state that the sum of weights will be more than the weight of one particular expert.

$$W_{n+1} \geq e^{-\beta L_{i,n}}$$

Upper Bound

Now, let us calculate the upper bound. We know from Algorithm 3 that –

$$\frac{W_{t+1}}{W_t} = \frac{\sum_{i=1}^K W_{i,t} \cdot e^{-\beta \ell_{i,t}}}{\sum_{i=1}^K W_{i,t}}$$

Let us now define a random variable Z such that –

$$Z = Y_{i,t} \quad \text{w. p.} \quad \frac{W_{i,t}}{\sum_{i=1}^K W_{i,t}}$$

In other words, the random variable Z takes the value of the expert with the probability of the algorithm choosing that expert. Hence, it is clear to see that $A_t = \mathbb{E}[Z]$. Also, we can see from the previous expression that the fraction $\frac{W_{t+1}}{W_t}$ is just a weighted sum of the exponent values and their probabilities. Hence, we can say –

$$\frac{W_{t+1}}{W_t} = \sum_{i=1}^K \frac{W_{i,t}}{\sum_{i=1}^K W_{i,t}} \cdot e^{-\beta \ell_{i,t}} = \sum_{i=1}^K p_t(i) \cdot e^{-\beta \ell_{i,t}} \mathbb{E} \left[e^{-\beta \cdot \ell(X_t, Z)} \right]$$

Applying Lemma 1.4, we get –

$$\frac{W_{t+1}}{W_t} = \mathbb{E} \left[e^{-\beta \cdot \ell(X_t, Z)} \right] \leq e^{-\beta \cdot \mathbb{E}[\ell(X_t, Z)]} \cdot e^{\frac{\beta^2}{8}}$$

Using Jensen's inequality (bcoz ℓ is convex in Z)

$$\mathbb{E}[\ell(X_t, Z)] \geq \ell(X_t, \mathbb{E}[Z]) = \ell(X_t, A_t)$$

Substituting, we get –

$$\frac{W_{t+1}}{W_t} \leq e^{-\beta \cdot \ell(X_t, A_t)} \cdot e^{\frac{\beta^2}{8}}$$

Now, we know that –

$$\frac{W_{n+1}}{W_1} = \frac{W_{n+1}}{K} = \prod_{t=1}^n \frac{W_{t+1}}{W_t}$$

Thus, we get –

$$e^{\beta L_{i,n}} \leq W_{n+1} \leq K \cdot e^{\frac{n\beta^2}{8}} \cdot e^{\beta L_n}$$

This gives us the final bound as –

$$\boxed{L_n \leq L_{i,n} + \frac{n\beta}{8} + \frac{\ln K}{\beta}} \tag{3}$$

The tightest upper bound will be when we take the best expert. Then –

$$\boxed{L_n^{\text{EWMA}} \leq \left[\min_i L_{i,n} \right] + \frac{n\beta}{8} + \frac{\ln K}{\beta}} \tag{4}$$

Now, if we set $\beta = \sqrt{\frac{8 \ln K}{n}}$, we get the regret as –

$$R_n^{\text{EWMA}} = L_n^{\text{EWMA}} - \min_i L_{i,n} \leq \sqrt{\frac{n \ln K}{2}}$$

Thus, we get a sub-linear regret bound with respect to n using exponential decay of weights and non-binary action. Effectively, we have relaxed the assumption that our loss is a binary loss and used a softer, convex proxy for the same.

Note

A caveat to note here is that if we wish to guarantee sub-linear regret, Algorithm 3 needs to be aware about the **horizon** n a priori. Thus, if n is not known, the algorithm fails to give sub-linear regret. An algorithm that can work without knowing n is called an **anytime algorithm** and unfortunately Algorithm 3 is not an anytime algorithm.

Question 1

Try this exercise. Suppose we try to make 3 as an anytime algorithm. In that case, we need to remove the dependence of β on n . This is done by setting β to be decaying with time i.e. β_t .

1. Try setting $\beta_t = \frac{1}{\sqrt{t}}$ and check if the regret would be sub-linear.
2. Try setting $\beta_t = \frac{1}{t^2}$ and check if the regret would be sub-linear.

Answer

Let us start with the case where we set $\beta_t = \frac{1}{\sqrt{t}}$. In this case, for some expert i , the loss after some n steps becomes -

$$W_{i,n+1} = \exp \left\{ - \sum_{t=1}^n \beta_t \ell(X_t, Y_{i,t}) \right\}$$

Thus, the overall sum of weights after n time steps can be given by -

$$W_{n+1} \geq \exp \left\{ - \sum_{t=1}^n \beta_t \ell(X_t, Y_{i,t}) \right\}$$

This is the lower bound. Now, let us look at the upper bound. We know that -

$$\frac{W_{t+1}}{W_t} = \mathbb{E} [\exp \{ -\beta_t \ell(X_t, Y_{i,t}) \}] \leq \exp \left\{ -\beta_t \mathbb{E} [\ell(X_t, Y_{i,t})] + \frac{\beta_t^2}{8} \right\}$$

Telescoping throughout the sequence,

$$\frac{W_{n+1}}{W_1} = \prod_{t=1}^n \frac{W_{t+1}}{W_t} \leq \exp \left\{ \sum_{t=1}^n -\beta_t \mathbb{E} [\ell(X_t, Y_{i,t})] + \frac{\beta_t^2}{8} \right\}$$

Combining the two, we get -

$$\exp \left\{ - \sum_{t=1}^n \beta_t \ell(X_t, Y_{i,t}) \right\} \leq W_{n+1} \leq K \cdot \exp \left\{ \sum_{t=1}^n -\beta_t \mathbb{E} [\ell(X_t, Y_{i,t})] + \frac{\beta_t^2}{8} \right\}$$

Substituting $\beta_t = \frac{1}{\sqrt{t}}$ and using sum of GP expression, we get the bounds. Similarly, we can substitute $\beta_t = \frac{1}{t^2}$ and get the bounds.

Randomized Exponential WMA

Assume that the values $Y_{i,t}, A_t \in \mathcal{D}$ where $\mathcal{D}$ is **convex**. The input X_t on the other hand belongs to set $\mathcal{E}$ which is arbitrary². Thus, we get a loss function $\ell: \mathcal{E} \times \mathcal{D} \rightarrow [0, 1]$ such that $\ell(x, \cdot)$ is convex $\forall x \in \mathcal{E}$.

In this case as well, this formulation behaves the same way as the previous formulation and thus we get -

$$R_n^{\text{EWMA}} \leq \sqrt{\frac{n \ln K}{2}} \quad ; \quad \beta = \sqrt{\frac{8 \log K}{n}}$$

²need not be convex.

Note

For WMA (any variation), we are taking averages of the actions of the experts. Hence, $Y_{i,t}, A_t$ need to belong to the same convex space. If we want them to belong to different spaces, we need to change the algorithm.

In this generalization, we get rid of the assumption that $\mathcal{D}$ is convex. At this level of generality, no deterministic algorithm can achieve sub-linear regret. So the only way to get sub-linear regret (and trump over the adversary) is to use a **randomized algorithm**.

Algorithm 4 Randomized Exponential WMA

Require: Number of expert K , horizon n , learning rate $\beta \in (0, 1)$

```

1: Initialize: Weight vector  $W_0 = \{1, 1, \dots, 1\}$ 
2: for  $t = 1$  to  $n$  do
3:   Choose an expert  $I_t$  with probability  $\mathbb{P}(I_t = i) = \frac{W_t[i]}{\sum_{j=1}^K W_t[j]}$   $\triangleright$  This is the randomization part
4:
5:    $A_t = Y_{I_t, t}$   $\triangleright$  This is valid since  $A_t, Y_{i,t}$  live in the same space
6:   Observe  $X_t$ 
7:   for  $i = 1$  to  $K$  do
8:      $W_{t+1}[i] = W_t[i]e^{-\beta \cdot \ell(X_t, Y_{i,t})}$ 
9:   end for
10: end for

```

In Algorithm 4, the weight update is the same as Algorithm 3, but choosing an expert is a random process. Now let us analyse the regret for Algorithm 4. Define -

$$D' = \left\{ p \in \mathbb{R}_+^K : \sum_{i=1}^K p_i = 1 \right\}$$

Essentially, D' is K -dimensional simplex. We also define -

$$E' = E \times D^K$$

where E is the space of X_t under Formulation 3. Now, we define $Y'_{i,t}$ as a vector of all zeros expect for the i^{th} position which will be 1. Thus, it gives us an idea which expert is being chosen. We also define -

$$X'_t = (X_t, [Y_{1,t}, \dots, Y_{K,t}])$$

Note that $X'_t \in E'$ and $Y'_{i,t} \in D'$. Now, we can proceed to formulate the loss function -

$$\ell'(x', p) = \ell((x, [y_1, \dots, y_K]), p) = \sum_{i=1}^K p_i \ell(x, y_i)$$

We need ℓ' to be convex w.r.t p which is true. Under Algorithm 3, we get -

$$A'_t = \left(\frac{w_{1,t}}{\sum_{j=1}^K w_{j,t}}, \dots, \frac{w_{K,t}}{\sum_{j=1}^K w_{j,t}} \right) \in D'$$

Also, verify that -

$$\ell'(X'_t, Y'_{i,t}) = \ell(X_t, Y_{i,t})$$

This is interesting because we can see that the loss for Algorithm 4 and Algorithm 3 are the **SAME** and implies that the weight update would be the same as well. Also, we have -

$$\ell'(X'_t, A'_t) = \sum_{i=1}^K \left(\frac{w_{i,t}}{\sum_{j=1}^K w_{j,t}} \right) \ell(X_t, Y_{i,t}) = \mathbb{E}[\ell(X_t, A_t)]$$

Thus, we have proven that the loss of Algorithm 4 is the expectation of the loss of Algorithm 3. Thus, the **expected regret** of Algorithm 4 is the same as the **regret** of Algorithm 3 and hence bounded as shown previously.

$$\mathbb{E} [R_n^{\text{REWMA}}] \leq \sqrt{\frac{n \ln K}{2}} \quad ; \quad \beta = \sqrt{\frac{8 \log K}{n}}$$

However, we have gotten the expected regret. But what if the actual regret deviates significantly from the expected regret? This deviation can be bound using **Hoeffding's inequality** -

$$\sum_{i=1}^n X_i \leq \mathbb{E} \left[\sum_{i=1}^n X_i \right] + \sqrt{\frac{n \ln \left(\frac{1}{\delta} \right)}{2}} \quad \text{w.p. } 1 - \delta \quad (5)$$

Claim 1.5. Under Algorithm 4 with $\beta = \sqrt{\frac{8 \ln K}{n}}$, we get regret bound as -

$$R_n^{\text{REWMA}} \leq \mathbb{E}[R_n^{\text{REWMA}}] + \sqrt{\frac{n \ln \left(\frac{1}{\delta} \right)}{2}} \leq \sqrt{\frac{n}{2}} \left(\sqrt{\frac{\ln K}{2}} + \sqrt{\frac{\ln \left(\frac{1}{\delta} \right)}{2}} \right)$$

Note

For applying Hoeffding's inequality, we need to have the random variables to be independent. Even though the distribution of A_t is dependent on the weights till $t - 1$, the random variables which are the loss values are still independent.

2 Multi-Armed Bandits (MAB)

In the mixture of experts setting, one very strong assumption we had considered was that at each time step, after we choose the action, we get to see what loss the other experts had experienced as well. However, what if we operate in a setting where after we choose an expert at time t , we see the loss of only the chosen expert and the other experts are hidden. What to do now?

Note

In the mixture of experts, we were observing losses of the other experts as well, which enabled us to exploit better. However, in MAB we are not sure which expert (also called **arm**) will be better than our current arm. Hence, this forces us to *explore* first and then try to exploit.

EXP3 [1]

This is an **Exponential** algorithm for **Exploration** and **Exploitation** (hence the name). At time t , we have the history vector $I_{t-1} = (A_1, Y_1, \dots, A_{t-1}, Y_{t-1})$ i.e. past information. Our task is to find A_t based on I_{t-1} . For an algorithm, we also define the following probability value -

$$p_{j,t} = \mathbb{P}[A_t = j | I_{t-1}]$$

which is the probability of choosing action j at time t given history I_{t-1} . We also compute an estimator -

$$\hat{Y}_{j,t} = \frac{\mathbb{1}_{A_t=j} \cdot Y_t}{p_{j,t}} = \begin{cases} \frac{Y_{j,t}}{p_{j,t}} & \text{for } j = A_t \\ 0 & \text{otherwise} \end{cases}$$

The above estimator is **unbiased**. This can be shown by checking -

$$\mathbb{E}[\hat{Y}_{j,t} | I_{t-1}] = \frac{p_{j,t} \cdot Y_{j,t}}{p_{j,t}} + \sum_{i \neq j} \frac{p_{i,t} \cdot 0}{p_{j,t}} = Y_{j,t}$$

Note

$Y_{j,t}$ is the loss of that the arm j experiences at time t . On the other hand, Y_t is the loss that the agent/algorithm experiences at time t after choosing action A_t i.e. $Y_t = Y_{A_t,t}$

However, there is an issue with the unbiased estimator. Let us denote

$$\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot | I_{t-1}]$$

Then, the conditional variance for the estimator can be given as -

$$\text{Var}(\hat{Y}_{j,t} | I_{t-1}) = \mathbb{E}_t[\hat{Y}_{j,t}^2] - (\mathbb{E}_t[\hat{Y}_{j,t}])^2 = \frac{Y_{j,t}^2 \cdot p_{j,t}}{p_{j,t}^2} - Y_{j,t}^2 = \frac{Y_{j,t}^2(1 - p_{j,t})}{p_{j,t}}$$

If the probability of choosing an arm is low and the reward is bounded away from 0, then the variance can be very large. In a similar sense, we can also define an estimator for the **losses** instead of the rewards as follows -

$$\hat{Y}_t = 1 - \frac{\mathbb{1}_{A_t=j} \cdot (1 - Y_t)}{p_{j,t}}$$

Setting $Z_t = 1 - Y_t$ and $\hat{Z}_t = 1 - \hat{Y}_t$ as the true and estimated losses, we get -

$$\hat{Z}_t = 1 - \frac{\mathbb{1}_{A_t=j} \cdot (1 - Z_t)}{p_{j,t}}$$

Now, we can also define the cumulative loss of an expert i upto time t as follows -

$$\hat{L}_{i,t} = \sum_{s=1}^t \hat{Z}_{i,s}$$

Algorithm 5 EXP3

Require: Number of arms K , horizon n , learning rate $\eta \in (0, 1)$

Initialize:

for $t = 1$ to n **do**

for $j = 1$ to K **do**

 Calculate $p_{j,t} = \frac{\exp(\eta \hat{L}_{j,t-1})}{\sum_{i=1}^K \exp(\eta \hat{L}_{i,t-1})}$

end for

 Sample $A_t \sim (p_{1,t}, \dots, p_{K,t})$

for $j = 1$ to K **do**

 Update $\hat{L}_{j,t} = \hat{L}_{j,t-1} + \hat{Y}_{j,t}$

end for

end for

Theorem 2.1. *The expected regret bound for 5 is given as -*

$$\mathbb{E} [R_n^{EXP3}] \leq \frac{\ln K}{\eta} + \eta n K$$

Setting $\eta = \sqrt{\frac{\ln K}{nK}}$, we get -

$$\mathbb{E} [R_n^{EXP3}] \leq 2\sqrt{nK \ln(K)}$$

Note

EXP3 is the same as 4 where instead of using the true loss L , we are using an estimation of the loss $\hat{L}$ at each time step.

Proof. We know that -

$$\mathbb{E}[\hat{L}_{i,t}] = \sum_{s=1}^t \mathbb{E}[\hat{Y}_{i,s}] = \sum_{s=1}^t \mathbb{E}[\mathbb{E}[\hat{Y}_{i,s} | I_{s-1}]] = \sum_{s=1}^t Y_{i,s} = L_{i,t}$$

Thus, the cumulative loss estimate is **unbiased**. We can also write -

$$\mathbb{E}[Y_t | I_{t-1}] = \sum_{j=1}^K p_{j,t} Y_{j,t}$$

We can also define the overall estimated loss over the horizon as -

$$\hat{L}_n = \sum_{t=1}^n \sum_{i=1}^K p_{i,t} \hat{Y}_{i,t}$$

Let us try to find the expectation of $\hat{L}_n$.

$$\begin{aligned} \mathbb{E}[\hat{L}_n] &= \sum_{t=1}^n \sum_{i=1}^K \mathbb{E}[\mathbb{E}[p_{i,t} \hat{Y}_{i,t} | I_{t-1}]] \\ &= \sum_{t=1}^n \sum_{i=1}^K \mathbb{E}[p_{i,t} \cdot \mathbb{E}[\hat{Y}_{i,t} | I_{t-1}]] \\ &= \sum_{t=1}^n \sum_{i=1}^K \mathbb{E}[p_{i,t} \cdot Y_{i,t}] \\ &= \sum_{t=1}^n \mathbb{E}[Y_t] \end{aligned}$$

Now, we can write -

$$\mathbb{E} [R_{i,n}^{\text{EXP3}}] = \sum_{t=1}^n \mathbb{E}[Y_t] - \sum_{t=1}^n Y_{i,t}$$

We shall define a weight value -

$$w_t = e^{\eta t} \sum_{i=1}^K e^{-\eta \hat{L}_{i,t}}$$

Thus, for an expert i ,

$$W_n \geq e^{\eta n} e^{-\eta \hat{L}_{i,n}}$$

We can also write -

$$\begin{aligned} \frac{W_t}{W_{t-1}} &= \frac{e^{\eta \sum_{j=1}^K e^{-\eta \hat{L}_{j,t-1}}} \cdot e^{-\eta \hat{Y}_{j,t}}}{\sum_{i=1}^K e^{-\eta \hat{L}_{i,t-1}}} \\ &= e^{\eta \sum_{j=1}^K p_{j,t} e^{-\eta \hat{Y}_{j,t}}} \\ &= \sum_{j=1}^K p_{j,t} e^{\eta(1-\hat{Y}_{j,t})} \end{aligned}$$

We know that for all $x \in (-\infty, 1]$, we have -

$$1 + x \leq e^x \leq 1 + x + x^2$$

Thus, we get -

$$\begin{aligned} \frac{W_t}{W_{t-1}} &= \sum_{j=1}^K p_{j,t} e^{\eta(1-\hat{Y}_{j,t})} \leq \sum_{j=1}^K p_{j,t} \left[1 + \eta(1-\hat{Y}_{j,t}) + \eta^2(1-\hat{Y}_{j,t})^2 \right] \\ &= 1 + \eta \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t}) + \eta^2 \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t})^2 \\ &\leq \exp \left\{ \eta \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t}) + \eta^2 \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t})^2 \right\} \end{aligned}$$

Thus, we can write -

$$\begin{aligned} \prod_{t=1}^n \frac{W_t}{W_{t-1}} &= \frac{W_n}{K} \leq \exp \left\{ \eta \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t}) + \eta^2 \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t})^2 \right\} \\ e^{\eta n} e^{-\eta \hat{L}_{i,n}} &\leq K \exp \left\{ \eta \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t}) + \eta^2 \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t})^2 \right\} \\ \eta n - \eta \hat{L}_{i,n} &\leq \ln K + \eta \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t}) + \eta^2 \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t})^2 \\ n - \hat{L}_{i,n} &\leq \frac{\ln K}{\eta} + \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t}) + \eta \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t})^2 \\ n - \hat{L}_{i,n} &\leq \frac{\ln K}{\eta} + n - \hat{L}_n + \eta \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t})^2 \\ \hat{L}_n - \hat{L}_{i,n} &\leq \frac{\ln K}{\eta} + \eta \sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1-\hat{Y}_{j,t})^2 \end{aligned}$$

We can now use Lemma A.1 to get -

$$\boxed{\mathbb{E} [R_{i,n}^{\text{EXP3}}] = \hat{L}_n - \hat{L}_{i,n} \leq \frac{\ln K}{\eta} + \eta K n}$$

□

Note

As compared to 4, the regret bound for 5 has an extra $\sqrt{\ln K}$ factor.

Note

Probability primer

Till now, we have seen the mixture of experts settings and bandits settings^a, which were both in an adversarial setting. Now, we shall transition into the cases where the sequences/rewards are drawn stochastically, rather than adversarially. Prior to proceeding, read up on the probability basics (refer B). We need to know about sub-Gaussian and sub-Exponential random variables.

^aIt is essentially a mixture of experts, but the losses of the other experts are not known at a time step.

3 Stochastic MABs

We consider a setting which has K arms and each arm $i \in [K]$ is associated with **reward distribution** ν_i . The MAB instance $\nu = [\nu_1, \dots, \nu_K]$ is the collection of all the reward distributions of the arms. At each time $t > 0$,

- The algorithm pulls an arm $A_t \in [K]$
- The algorithm receives a reward $X_t \sim \nu_{A_t}$

At the end of the horizon n , our goal is to maximize the expected reward.

Intuition

We have no idea about the underlying arm distributions. However, there is an **oracle** that knows all the distributions. In expectation, how bad do we perform as compared to the oracle? This is the **regret**.

Assume that μ_i represents the mean reward of arm i . Also, define -

$$\mu^* = \max_{i \in [K]} \mu_i$$

Then, over an horizon of n , we can write the regret as -

$$R_n(\pi, \nu) = n\mu^* - \mathbb{E} \left[\sum_{t=1}^n X_t \right]$$

Our goal is to minimize R_n . Let us define a **sub-optimality gap** of an arm i as -

$$\Delta_i \stackrel{\text{def}}{=} \mu^* - \mu_i$$

Also, we use $T_i(t)$ to represent the number of times arm i was pulled till time t . Using these two quantities, we can re-define the regret as shown below.

Lemma 3.1. *We can re-define the regret as -*

$$R_n(\pi, \nu) = \sum_{i \in [K]} \Delta_i \cdot \mathbb{E} [T_i(n)]$$

Proof. We know that

$$R_n = \sum_{t=1}^n \mathbb{E} [\mu^* - \mu_{A_t}] = \sum_{t=1}^n \mathbb{E} [\Delta_{A_t}]$$

Additionally, we know that

$$\Delta_{A_t} = \sum_{i=1}^K \mathbb{1}[A_t = i] \cdot \Delta_i$$

Plus, we know that

$$T_i(n) = \sum_{t=1}^n \mathbb{1}[A_t = i]$$

Combining, we get -

$$R_n = \sum_{t=1}^n \mathbb{E} \left[\sum_{i=1}^K \mathbb{1}[A_t = i] \cdot \Delta_i \right] = \sum_{i=1}^K \Delta_i \cdot \mathbb{E} \left[\sum_{t=1}^n \mathbb{1}[A_t = i] \right] = \sum_{i=1}^K \Delta_i \cdot \mathbb{E} [T_i(n)]$$

□

Note

We do not make an assumption that the optimal arm is unique. All the arms could have the highest expected reward and the framework still works.

Full-Info setting

Prior to understanding the actual MAB setting, we have a brief look at the Full-Info setting i.e. we observe the rewards of all the arms but experience the reward of the arm that was pulled. Essentially, at some time t , we have received t i.i.d samples of the rewards of each arm. Hence, calculating the empirical mean estimation and choosing the one with the max mean should work in our favour. This works since this is a benign setting and not an adversarial setting.

Algorithm 6 FullInfo setting

Require: Number of arms K , horizon n

Initialize: Pull any arm and observe the rewards of all arms.

for $t = 2$ to n **do**

for $j = 1$ to K **do**

 Calculate empirical mean $\hat{\mu}_j(t-1)$

end for

 Pick arm $A_t \in \arg \max_{i \in [K]} \hat{\mu}_i(t-1)$

end for

One thing to note here is that in Algorithm 6, if suppose we pulled arm 1 initially, then -

$$\begin{aligned} A_t = i &\implies \hat{\mu}_i(t-1) \geq \hat{\mu}_1(t-1) \\ \mathbb{P}(A_t = i) &\leq \mathbb{P}(\hat{\mu}_i(t-1) \geq \hat{\mu}_1(t-1)) \end{aligned}$$

Note

We are under the assumption that the arm rewards are 1-sub Gaussian. The reasonability of this assumption shall be explored later.

Thus, we get -

$$\mathbb{P}(A_t = i) \leq \mathbb{P}(\hat{\mu}_1(t-1) - \hat{\mu}_i(t-1) - \Delta_i \leq -\Delta_i)$$

Let us define a random variable - $X = \hat{\mu}_1(t-1) - \hat{\mu}_i(t-1)$. Then, we can say that $\mathbb{E}[X] = \Delta_i$.

Lemma 3.2. $X = \hat{\mu}_1(t-1) - \hat{\mu}_i(t-1)$ is a $\sqrt{\frac{2}{t-1}}$ -sub Gaussian.

Proof. Consider two random variables - A, B that are σ_1, σ_2 subGaussian respectively. After some t time steps, we can say that -

$$\sum_{i=1}^t A_i \sim \sqrt{t}\sigma_1 - \text{subG} \quad ; \quad \sum_{i=1}^t B_i \sim \sqrt{t}\sigma_2 - \text{subG}$$

This is assuming that each instance of the random variable is independent. Taking the average of these, we get -

$$\hat{\mu}_A(t) \sim \frac{\sigma_1}{\sqrt{t}} - \text{subG} \quad ; \quad \hat{\mu}_B(t) \sim \frac{\sigma_2}{\sqrt{t}} - \text{subG}$$

Now, we can write that the sum of two subG RVs with parameter σ_1, σ_2 is also a subG with parameter $\sqrt{\sigma_1^2 + \sigma_2^2}$. Thus, we can write -

$$\hat{\mu}_A(t) - \hat{\mu}_B(t) \sim \sqrt{\frac{\sigma_1^2 + \sigma_2^2}{t}} - \text{subG}$$

Since we are assuming the arms are 1-subG, we get $\sigma_1 = \sigma_2 = 1$. Substituting finishes the proof. $\square$

Then, using sub-Gaussian concentration, we get -

$$\mathbb{P}(A_t = i) \leq \mathbb{P}(X - \mathbb{E}[X] \leq -\Delta_i) \leq e^{-\frac{\Delta_i^2(t-1)}{4}}$$

Thus, as t goes on, the probability of picking a sub-optimal arm i ($\Delta_i > 0$) will decrease exponentially. Thus, we get -

$$\mathbb{E}[T_i(n)] \leq 1 + \sum_{t=2}^n \mathbb{P}(A_t = i) \leq 1 + \sum_{t=2}^n e^{-\frac{\Delta_i^2(t-1)}{4}} = O(1)$$

Thus, for the **Full-info setting** as described in Algorithm 6, the regret is a constant $O(1)$.

Bandit setting

This is not as simple as the Full-Info setting since we don't have i.i.d samples for each arm. Thus, in the Full-Info setting, we were just performing **exploitation**. However, in the bandit setting, we have to have a balance between **exploration and exploitation**.

One of the most natural algorithms is the **Explore-then-Commit (ETC)**.

Algorithm 7 ETC bandit algorithm

Require: Number of arms K , horizon n , number of pulls of each arm m

Initialize: Pull each arm m times.

for $j = 1$ to K **do**

 Calculate empirical mean $\hat{\mu}_j(m)$

end for

Pick arm $A \in \arg \max_{i \in [K]} \hat{\mu}_i(m)$

for $t = 2$ to n **do**

 Pull arm A

end for

Note

Ek baar maine committment kar di, toh main apni bhi nahi sunta.

Let us now check the regret bound of Algorithm 7. Let us assume WLOG that arm 1 is optimal. For each arm i , we get -

$$\mathbb{E}[T_i(n)] = m + \mathbb{P}(\hat{\mu}_i(m) \in \arg \max_{j \in [K]} \hat{\mu}_j(m)) \cdot (n - mK) = m + \mathbb{P}(\hat{\mu}_i(m) \in \hat{\mu}_1(m)) \cdot (n - mK)$$

Using the same process as the regret bound in Full-Info setting, we can write -

$$\mathbb{E}[T_i(n)] \leq m + (n - mK)e^{\frac{-\Delta_i^2(m)}{4}}$$

Thus, we get the regret bound as -

$$R_n^{\text{ETC}} \leq m \sum_{i=1}^K \Delta_i + (n - mK) \sum_{i=1}^K \Delta_i e^{\frac{-\Delta_i^2(m)}{4}}$$

Note

The regret bound is an **instance dependent** bound i.e. it is NOT agnostic to the probability distributions of the arms. Later on, we shall study instance independent bounds that give a bound irrespective of the probability distributions of the arms.

The next obvious question is whether how should we choose the value of $\mathbf{m}$? We can differentiate the expression for R_n^{ETC} and set it to zero to get m . But then, the value of m will be in terms of Δ_i which we don't know. So, we need another way to choose m . Let us consider a toy instance where we have 2 arms such that -

$$\Delta_1 = 0 \quad ; \quad \Delta_2 = \Delta > 0$$

Then we write -

$$R_n^{\text{ETC}} \leq m\Delta + (n - 2m)\Delta e^{\frac{-m\Delta^2}{4}}$$

This gives us -

$$m^* = \frac{4}{\Delta^2} \ln \left(\frac{n\Delta^2}{4} \right)$$

Which gives us the regret bound as -

$$R_n^{\text{ETC}} \leq \frac{4}{\Delta} \left[1 + \ln \left(\frac{n\Delta^2}{4} \right) \right]$$

Intuition

As the value of Δ becomes smaller (approaches 0), the $\frac{4}{\Delta^2}$ dominates and hence the value of m^* explodes. Thus, as the arms get closer in distribution, we need more time to figure out which arm is sub-optimal with high confidence.

Note

One more thing to note here is that even if we knew the value of Δ , we are still experiencing a regret that scaled as $O(\ln(n))$. This is a key observation i.e. even for the best case scenarios in the bandit settings, we achieve logarithmic regret.

Remarks

- To see logarithmic regret, having a lower bound on the value of Δ is enough. We don't really need the exact value of Δ .
- 7 is an anytime algorithm i.e. it doesn't depend on the value of n . However, finding m^* is dependent on n . We can use **doubling trick** to make this also anytime.
- Can we get logarithmic regret guarantees using m that is instance agnostic i.e. not dependent on Δ ?

Upper Confidence Bound (UCB)

This is a popular algorithm to achieve logarithmic regret and also has instance adaptive way of doing the exploration-exploitation tradeoff. Let us denote $\hat{\mu}_{i,n}$ be the empirical mean of arm i after n samples. Then, if the true mean for arm i is μ_i , then -

$$\mathbb{P}(\hat{\mu}_{i,n} - \mu_i \leq -\epsilon) \leq e^{\frac{-n\epsilon^2}{2}} = \delta \quad (\text{say})$$

This holds since we have assumed that the arms are **1-subGaussian**. Thus, we get

$$\epsilon = \sqrt{\frac{2 \ln\left(\frac{1}{\delta}\right)}{n}}$$

Which enables us to write -

$$\mu_i \leq \hat{\mu}_{i,n} + \sqrt{\frac{2 \ln\left(\frac{1}{\delta}\right)}{n}} \quad \text{w.p.} \geq 1 - \delta$$

The above bound is called the **UCB** and is used in the algorithm. We can define the UCB of arm i after time $t - 1$ as follows -

$$\text{UCB}_i(t-1) = \begin{cases} \infty & \text{if } T_i(t-1) = 0 \\ \hat{\mu}_{i,t-1} + \sqrt{\frac{2 \ln\left(\frac{1}{\delta}\right)}{T_i(t-1)}} & \text{otherwise} \end{cases}$$

Algorithm 8 UCB algorithm

Require: Number of arms K , horizon n , confidence parameter δ

Initialize: UCB of each arm as ∞ .

Pull each arm once.

for $t = K + 1$ to n **do**

for $j = 1$ to K **do**

 Update $\text{UCB}_j(t) = \hat{\mu}_{j,t-1} + \sqrt{\frac{2 \ln\left(\frac{1}{\delta}\right)}{T_j(t-1)}}$

end for

 Pick arm $A \in \arg \max_{i \in [K]} \text{UCB}_i(t-1)$

end for

Note

We assign an index to each arm and perform calculations independently (decentralization). These type of algorithms are often called **index algorithms**.

Theorem 3.3. Under the UCB algorithm as illustrated in 8, if we set $\delta = \frac{1}{n^2}$, we get logarithmic regret.

$$R_n \leq 3 \sum_{i=1}^K \Delta_i + \sum_{i: \Delta_i > 0} \frac{16 \log n}{\Delta_i}$$

Note that the second term only considers the sub-optimal arms.

Proof. Assume that WLOG Consider some sub-optimal arm i and define the following event -

$$G_i \stackrel{\text{def}}{=} \left\{ \mu_1 < \min_{t \in [n]} \text{UCB}_1(t) \right\} \cap \left\{ \hat{\mu}_{i, u_i} + \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{u_i}} < \mu_1 \right\}$$

This is a “good event” which consists of two events -

- First event says that Arm 1 is always valid.
- Second term says that the UCB of sub-optimal arm i is lesser than the true mean of the optimal arm 1

We first claim that $T_i(n) \leq u_i$ i.e. the total number of pulls of arm i up until time n is less than u_i . Suppose this statement is not true. Then, it would imply that arm i got u_i number of pulls and had at least one more time step t where UCB of arm i was highest. Formally -

$$\exists t < n \quad \text{s.t.} \quad T_i(t-1) \quad \text{AND} \quad A_t = i$$

We can say that if we have a good event G_i , then -

$$\text{UCB}_i(t) = \hat{\mu}_{i, u_i} + \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{u_i}} < \mu_1 < \text{UCB}_1(t)$$

This is a clear contradiction since UCB of arm 1 will be the highest and arm i shouldn't have been chosen. Thus, if a “good event” occurs, then $T_i(n) \leq u_i$. Now, we can write -

$$\mathbb{E}[T_i(n)] = \{\mathbb{P}(G_i) \cdot \mathbb{E}[T_i(n) \mid G_i]\} + \{\mathbb{P}(G_i^C) \cdot \mathbb{E}[T_i(n) \mid G_i^C]\} \leq u_i + \mathbb{P}(G_i^C) \cdot n$$

We know from the definition of G_i , we get -

$$G_i^C \stackrel{\text{def}}{=} \left\{ \mu_1 \geq \min_{t \in [n]} \text{UCB}_1(t) \right\} \cup \left\{ \hat{\mu}_{i, u_i} + \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{u_i}} \geq \mu_1 \right\} = J_1 \cup J_2$$

Let us consider the event J_1 which is essentially the event where the UCB of arm 1 drops below the true mean (this shouldn't happen since UCB stands for **U**pper confidence bound).

$$J_1 \implies \left\{ \mu_1 \geq \min_{S \in [n]} \hat{\mu}_{1, S} + \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{S}} \right\}$$

J_1 is the event that in the UCB run, arm 1 had a UCB lesser than the true mean. The RHS on the expression above is the event that if we pull the arm 1 from 1 to n times, there will be some time step S

where the UCB will be lesser than the true mean. Thus, it is clear to see that J_1 implies the RHS event. Hence, -

$$\begin{aligned}\mathbb{P}(J_1) &\leq \mathbb{P} \left[\left\{ \mu_1 \geq \min_{S \in [n]} \hat{\mu}_{1,S} + \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{S}} \right\} \right] \\ \mathbb{P}(J_1) &\leq \mathbb{P} \left[\bigcup_{s=1}^n \mu_1 \geq \hat{\mu}_{1,S} + \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{S}} \right] \\ \mathbb{P}(J_1) &\leq \sum_{s=1}^n \mathbb{P} \left[\mu_1 \geq \hat{\mu}_{1,S} + \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{S}} \right] \\ \mathbb{P}(J_1) &\leq n\delta\end{aligned}$$

Now, consider the event J_2 .

$$\begin{aligned}\mathbb{P}(J_2) &= \mathbb{P} \left[\left\{ \hat{\mu}_{i,u_i} + \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{u_i}} \geq \mu_1 \right\} \right] \\ \mathbb{P}(J_2) &= \mathbb{P} \left[\left\{ \hat{\mu}_{i,u_i} - \mu_i \geq \Delta_i - \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{u_i}} \right\} \right]\end{aligned}$$

Let us set

$$u_i = \left\lceil \frac{8 \ln \left(\frac{1}{\delta} \right)}{\Delta_i^2} \right\rceil$$

such that -

$$\Delta_i - \sqrt{\frac{2 \ln \left(\frac{1}{\delta} \right)}{u_i}} \geq \frac{\Delta_i}{2}$$

Then for this choice of u_i , we can write -

$$\mathbb{P}(J_2) \leq \mathbb{P} \left[\hat{\mu}_{i,u_i} - \mu_i \geq \frac{\Delta_i}{2} \right] \leq \exp \left[\frac{-u_i \Delta_i^2}{8} \right] = \delta$$

Substituting all of these back, we get -

$$\begin{aligned}\mathbb{E}[T_i(n)] &\leq u_i + \mathbb{P}(G_i^C) \cdot n \\ \mathbb{E}[T_i(n)] &\leq \left\lceil \frac{8 \ln \left(\frac{1}{\delta} \right)}{\Delta_i^2} \right\rceil + \mathbb{P}(J_1 \cup J_2) \cdot n \\ \mathbb{E}[T_i(n)] &\leq 1 + \frac{8 \ln \left(\frac{1}{\delta} \right)}{\Delta_i^2} + n (\mathbb{P}(J_1) + \mathbb{P}(J_2)) \\ \mathbb{E}[T_i(n)] &\leq 1 + \frac{8 \ln \left(\frac{1}{\delta} \right)}{\Delta_i^2} + n (n\delta + \delta)\end{aligned}$$

Setting $\delta = \frac{1}{n^2}$, we get -

$$\boxed{\mathbb{E}[T_i(n)] \leq 3 + \frac{16 \log n}{\Delta_i^2}}$$

We know that -

$$R_n = \sum_{i=1}^K \Delta_i \mathbb{E}[T_i(n)]$$

Substituting, we get the require expression.

$$R_n \leq 3 \sum_{i=1}^K \Delta_i + \sum_{i: \Delta_i > 0} \frac{16 \log n}{\Delta_i}$$

□

The bound above is an instance dependent bound i.e. depends on the instance we have defined. This is useful if the instance is kept fixed and n increases asymptotically. Now suppose we have a fixed horizon and the value of δ_i for some arm i is very close to 0. Then, the expected number of pulls blows up to ∞ which makes no sense since n is finite and you can't pull more than n times. Hence, we need an **instance independent** bound.

Theorem 3.4. *Under the UCB algorithm as illustrated in 8, if we set $\delta = \frac{1}{n^2}$, we get logarithmic regret.*

$$R_n \leq 8\sqrt{nK \ln(n)} + 3 \sum_{i=1}^K \Delta_i$$

Proof. Let us re-write the regret as follows -

$$R_n = \sum_{i=1}^K \Delta_i \mathbb{E}[T_i(n)] = \sum_{i: \Delta_i < \Delta} \Delta_i \mathbb{E}[T_i(n)] + \sum_{i: \Delta_i > \Delta} \Delta_i \mathbb{E}[T_i(n)]$$

For $\Delta_i > \Delta$, we can stick to the instance dependent bound itself.

$$\sum_{i: \Delta_i > \Delta} \Delta_i \mathbb{E}[T_i(n)] = 3 \sum_{i: \Delta_i > \Delta} \Delta_i + \sum_{i: \Delta_i > \Delta} \frac{16 \ln(n)}{\Delta_i} \leq 3 \sum_{i: \Delta_i > \Delta} \Delta_i + \frac{16K \ln(n)}{\Delta}$$

Thus, we get -

$$R_n \leq n\Delta + 3 \sum_{i: \Delta_i > \Delta} \Delta_i + \frac{16K \ln(n)}{\Delta}$$

We set $\Delta = \sqrt{\frac{16K \ln(n)}{n}}$ to get the tightest(required) bound.

$$R_n \leq 8\sqrt{nK \ln(n)} + 3 \sum_{i=1}^K \Delta_i$$

□

In the above bound, the dominant term $8\sqrt{nK \ln(n)}$ is independent of the arm instances.

Anytime UCB

In the previous iteration of the vanilla UCB, the algorithm as illustrated in 8 is **NOT anytime** i.e. it needs to know the value of horizon n from the start itself. We shall now explore the edition of UCB that is in fact anytime i.e. independent of n . The key idea here is a **decaying** value of δ .

$$\begin{aligned} \text{In vanilla UCB} &\longrightarrow \delta = \frac{1}{n^2} \\ \text{In anytime UCB} &\longrightarrow \delta = \frac{1}{t^4} \end{aligned}$$

We see that in 9, the value of UCB is independent of n . Let us now analyse the regret for this algorithm.

Algorithm 9 Anytime UCB algorithm

Require: Number of arms K , horizon n , confidence parameter δ

Initialize: UCB of each arm as ∞ .

Pull each arm once.

for $t = K + 1$ to n **do**

for $j = 1$ to K **do**

 Update $\text{UCB}_j(t) = \hat{\mu}_{j,t-1} + \sqrt{\frac{8 \ln(t)}{T_j(t-1)}}$

end for

 Pick arm $A \in \arg \max_{i \in [K]} \text{UCB}_i(t-1)$

end for

Theorem 3.5. *The regret bound on 9.*

Proof. Let us assume that arm i is sub-optimal and arm 1 is optimal (WLOG). We start off with a **claim** that if arm i was pulled at time t i.e. $A_t = i$, then one of three conditions should have occurred -

1. $\text{UCB}_1(t) \leq \mu_1$
2. $\text{LCB}_i(t) > \mu_i$
3. $T_i(t-1) < \frac{32 \ln(t)}{\Delta_i^2} \iff \frac{\Delta_i}{2} < \sqrt{\frac{8 \ln(t)}{T_i(t-1)}}$

The first condition corresponds to the fact that arm 1 has performed poorly i.e. sampled values are all lower than the mean so far. On the other hand, the condition corresponds to the fact that arm i has performed exceptionally well i.e. sampled values are all higher than the mean so far. The last condition relates to the fact that arm i hasn't been pulled too often.

Intuitively, the three conditions make sense. In either scenario, arm i which is sub-optimal seems to be a good option to pull. However, we must prove the claim that having either one of these three is sufficient to cause arm i to be pulled.

Intuition

We shall prove the negation. As per the claim, arm i getting pulled implies one of the three holds. We shall prove that if none of the three conditions hold, arm i will not hold.

The negation of the three conditions are as follows -

1. $\text{UCB}_1(t) > \mu_1$
2. $\text{LCB}_i(t) \leq \mu_i$
3. $T_i(t-1) \geq \frac{32 \ln(t)}{\Delta_i^2} \iff \frac{\Delta_i}{2} \geq \sqrt{\frac{8 \ln(t)}{T_i(t-1)}}$

We know that -

$$\text{UCB}_1(t) > \mu_1 = \mu_i + \Delta_i \geq \text{LCB}_i(t) + \Delta_i$$

These follow from the first two (negated) claims. From claim (negated) 3, we can write -

$$\text{UCB}_1(t) > \hat{\mu}_i(t) - \sqrt{\frac{8 \ln(t)}{T_i(t-1)}} + 2\sqrt{\frac{8 \ln(t)}{T_i(t-1)}} = \hat{\mu}_i(t) + \sqrt{\frac{8 \ln(t)}{T_i(t-1)}} = \text{UCB}_i(t)$$

In other words, if none of the conditions hold, then arm i will **never** be pulled.

So, we have proven the claim with the three conditions. Let us now use that to get the regret bound. Consider -

$$\begin{aligned} T_i(n) &= \sum_{t=1}^n \mathbb{1} \{A_t = i\} \\ &\leq \sum_{t=1}^n \mathbb{1} \{A_t = i \& \text{Condition 1 holds}\} + \sum_{t=1}^n \mathbb{1} \{A_t = i \& \text{Condition 2 holds}\} + \sum_{t=1}^n \mathbb{1} \{A_t = i \& \text{Condition 3 holds}\} \end{aligned}$$

This follows from the fact that pulling arm i happens when one of the three conditions hold. Let us consider the third summation. Then, we get -

$$\sum_{t=1}^n \mathbb{1} \{A_t = i \& \text{Condition 3 holds}\} \leq 1 + \frac{32 \ln(n)}{\Delta_i^2}$$

Hence, we get -

$$T_i(n) \leq 1 + \frac{32 \ln(n)}{\Delta_i^2} + \sum_{t=1}^n \mathbb{1} \{A_t = i \& \text{Condition 1 holds}\} + \sum_{t=1}^n \mathbb{1} \{A_t = i \& \text{Condition 2 holds}\}$$

Taking expectation on both sides -

$$\mathbb{E}[T_i(n)] \leq 1 + \frac{32 \ln(n)}{\Delta_i^2} + \sum_{t=1}^n \mathbb{P} \{\text{Condition 1 holds}\} + \sum_{t=1}^n \mathbb{P} \{\text{Condition 2 holds}\}$$

Let us take the first probability.

$$\mathbb{P} \{\text{Condition 1 holds}\} = \mathbb{P} \{\text{UCB}_1(t) \leq \mu_1\}$$

We claim that -

$$\text{UCB}_1(t) \leq \mu_1 \implies \bigcup_{s=1}^t \{\text{UCB}_1(s) \leq \mu_1\}$$

The argument in RHS is the condition that UCB is violated when arm 1 gets s pulls. If we take a union for s from 1 to t , we get the event that UCB violation happens if the arm 1 gets pulled either 1 times, 2 times, all the way to t times. Hence, a violation at time t implies the union of the events.

$$\text{UCB}_1(t) \leq \mu_1 \implies \bigcup_{s=1}^t \left\{ \hat{\mu}_1(s) + \sqrt{\frac{8 \ln(t)}{s}} \leq \mu_1 \right\}$$

Applying union bound helps us get the probability.

$$\mathbb{P} \{\text{UCB}_1(t) \leq \mu_1\} \leq \mathbb{P} \left\{ \bigcup_{s=1}^t \left[\hat{\mu}_1(s) + \sqrt{\frac{8 \ln(t)}{s}} \leq \mu_1 \right] \right\} \leq \sum_{s=1}^t \mathbb{P} \left[\hat{\mu}_1(s) + \sqrt{\frac{8 \ln(t)}{s}} \leq \mu_1 \right]$$

We know that the violation probability for each of the s timesteps is upper-bound by δ . Hence, we can write -

$$\mathbb{P} \{\text{UCB}_1(t) \leq \mu_1\} \leq \sum_{s=1}^t \delta = t \cdot \frac{1}{t^4}$$

Which gives us -

$$\boxed{\mathbb{P} \{\text{UCB}_1(t) \leq \mu_1\} \leq t^{-3}}$$

Using similar arguments, we can write -

$$\boxed{\mathbb{P} \{\text{Condition 2 holds}\} = \mathbb{P} \{\text{LCB}_1(t) > \mu_1\} \leq t^{-3}}$$

Hence, we can write -

$$\begin{aligned}
T_i(n) &\leq 1 + \frac{32 \ln(n)}{\Delta_i^2} + \sum_{t=1}^n t^{-3} + \sum_{t=1}^n t^{-3} \\
&\leq 1 + \frac{32 \ln(n)}{\Delta_i^2} + 2 \cdot \sum_{t=1}^{\infty} t^{-3} \\
&\leq 1 + \frac{32 \ln(n)}{\Delta_i^2} + 2\zeta(3)
\end{aligned}$$

Where $\zeta(3)$ is the sum of t^{-3} to ∞ , which is a constant called **Apéry's constant**. Hence, the overall regret becomes -

$$R_n(t) \leq (1 + 2\zeta(3)) \sum_{i=1}^K \Delta_i + \sum_{i: \Delta_i > 0} \frac{32 \ln(n)}{\Delta_i}$$

□

Note

We can see that the regret for 9 is higher than 8, which is expected from anytime algorithm. However, can we do better?

Please check that if we use $\delta = \frac{1}{t^3}$, then we get -

$$R_n(t) \leq \left(1 + \frac{\pi^2}{3}\right) \sum_{i=1}^K \Delta_i + \sum_{i: \Delta_i > 0} \frac{24 \ln(n)}{\Delta_i}$$

So we get a higher value for term 1 but a considerably smaller value for term 2. In other words, we can take any $\delta_t = t^{-\alpha}$ with $\alpha > 2$ and the resulting algorithm we get is α -UCB.

ϵ -Greedy

Algorithm 10 ϵ -Greedy algorithm

Require: Number of arms K , horizon n , ϵ_0 bias

Initialize: Empirical mean of each arm as 0.

for $t = 1$ to n **do**

Pick arm $A = \begin{cases} \text{Random} & \text{w.p. } \epsilon_t \\ \arg \max_i \hat{\mu}_i & \text{w.p. } 1 - \epsilon_t \end{cases}$

Decay $\epsilon_t = \frac{1}{t}$

end for

4 Information Theoretic Lower Bounds

B-H inequality

Theorem 4.1. If $\mathbb{P}$ and $\mathbb{Q}$ are probability measures on $(\Omega, \mathcal{F})$ such that they are absolutely continuous with respect to **each other**, then for any event $A \in \mathcal{F}$,

$$\mathbb{P}(A) + \mathbb{Q}(A^C) \geq \frac{1}{2} \exp \{ - \min [KL(\mathbb{P}, \mathbb{Q}), KL(\mathbb{Q}, \mathbb{P})] \}$$

Intuitively speaking, the above inequality states that if the two measures are “close” to each other, then the LHS (sum of probability of an event and its complement) can’t be too small. When $\mathbb{P} = \mathbb{Q}$, then -

$$\begin{aligned}\text{LHS} &= \mathbb{P}(A) + \mathbb{Q}(A^C) = \mathbb{P}(A) + \mathbb{P}(A^C) = 1 \\ \text{RHS} &= \frac{1}{2} \exp \{ -\min [\text{KL}(\mathbb{P}, \mathbb{Q}), \text{KL}(\mathbb{Q}, \mathbb{P})] \} = \frac{1}{2}\end{aligned}$$

We can see that $1 \geq \frac{1}{2}$, indicating that the inequality is satisfied.

Intuition

If an event is rare in $\mathbb{P}$, then it has to be rare in $\mathbb{Q}$ as well if the measures are close (implying that the event’s complement is not rare in $\mathbb{Q}$).

Proof. Consider a finite probability space where -

$$\Omega = \{x_1, \dots, x_n\} \quad ; \quad \mathcal{F} = \mathcal{P}(\Omega)$$

where we define two measures— $\mathbb{P}$ and $\mathbb{Q}$ —such that -

$$\begin{aligned}\mathbb{P}(\{x_i\}) &= p_i \quad \forall i \quad \text{s.t.} \quad p_i > 0 \ \& \ \sum_{i=1}^n p_i = 1 \quad \forall i \\ \mathbb{Q}(\{x_i\}) &= q_i \quad \forall i \quad \text{s.t.} \quad q_i > 0 \ \& \ \sum_{i=1}^n q_i = 1 \quad \forall i\end{aligned}$$

Also assume that $\mathbb{P}$ is absolutely continuous w.r.t $\mathbb{Q}$. Note that the LHS and RHS of the inequality are symmetric in $\mathbb{P}$ and $\mathbb{Q}$, which implies that proving the case for $\mathbb{P}$ being absolutely continuous w.r.t $\mathbb{Q}$ is sufficient and a symmetric argument can be made for the case when $\mathbb{Q}$ is absolutely continuous w.r.t $\mathbb{P}$.

Step 1

We first show that -

$$\begin{aligned}\mathbb{P}(A) + \mathbb{Q}(A^C) &= \sum_{x_i \in A} p_i + \sum_{x_i \in A^c} q_i \\ &\geq \sum_{i \in [n]} \min(p_i, q_i)\end{aligned}$$

Step 2

We show that -

$$\begin{aligned}\left(\sum_{i \in [n]} \sqrt{p_i q_i} \right)^2 &= \left(\sum_{i \in [n]} \sqrt{\min(p_i, q_i) \cdot \max(p_i, q_i)} \right)^2 \\ &\leq \left[\sum_{i \in [n]} \sqrt{\min(p_i, q_i)} \right] \cdot \left[\sum_{i \in [n]} \sqrt{\max(p_i, q_i)} \right] \quad \text{Cauchy-Schwartz inequality} \\ &\leq 2 \left[\sum_{i \in [n]} \sqrt{\min(p_i, q_i)} \right]\end{aligned}$$

Thus, we can write -

$$\sum_{i \in [n]} \sqrt{\min(p_i, q_i)} \geq \frac{1}{2} \left(\sum_{i \in [n]} \sqrt{p_i q_i} \right)^2$$

Step 3

We show that -

$$\begin{aligned}
\left(\sum_{i \in [n]} \sqrt{p_i q_i} \right)^2 &= \exp \left\{ 2 \log \left[\sum_{i \in [n]} \sqrt{p_i q_i} \right] \right\} \\
&= \exp \left\{ 2 \log \left[\sum_{i \in [n]} p_i \sqrt{\frac{q_i}{p_i}} \right] \right\} \\
&\geq \exp \left\{ 2 \sum_{i \in [n]} p_i \log \left[\sqrt{\frac{q_i}{p_i}} \right] \right\} \quad \text{Using Jensen's inequality} \\
&= \exp \left\{ - \sum_{i \in [n]} p_i \log \left[\frac{p_i}{q_i} \right] \right\}
\end{aligned}$$

Thus, we can write -

$$\left(\sum_{i \in [n]} \sqrt{p_i q_i} \right)^2 \geq \exp \left\{ - \sum_{i \in [n]} p_i \log \left[\frac{p_i}{q_i} \right] \right\} = \exp \{ -\text{KL}(\mathbb{P}, \mathbb{Q}) \}$$

Combining the results from the three steps, we get the required expression. $\square$

BH inequality

Let $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures defined on the same probability space and let A be an event. Then,

$$\mathbb{P}(A) + \mathbb{Q}(A^C) \geq \frac{1}{2} \exp \{ -\text{KL}(\mathbb{P}, \mathbb{Q}) \}$$

This inequality says that event A on $\mathbb{P}$ and A^C on $\mathbb{Q}$ can't be rare at the same time.

Proof. TBFL $\square$

Divergence-Decomposition Lemma

Lemma 4.2. $KL(\mathbb{P}_\mu, \mathbb{P}_{\mu'}) = \sum_{i=1}^K \mathbb{E}_\mu[T_i(n)] \cdot KL(\mu, \mu')$

Now we can proceed with the finding of lower bound. Let us consider $\mathcal{E}$ to be the space of the arm distributions under consideration. So far, we had been assuming $\mathcal{E}$ to be the family of 1-subGaussian random variables. We can respectively denote $\mathcal{E}^K$ as the family of MAB instances. For an arm-selection policy Π and a bandit instance ν , we define the regret over the horizon n as $R_n(\Pi, \nu)$.

$$R_n(\Pi, \mathcal{E}) = \sup_{\nu \in \mathcal{E}^K} R_n(\Pi, \nu)$$

The above is the worst-case regret we can have for the family of MABs considering the policy Π and fixed horizon n . Similarly, we can define -

$$R_n^*(\mathcal{E}) = \inf_{\Pi} R_n(\Pi, \mathcal{E}) = \inf_{\Pi} \sup_{\nu \in \mathcal{E}^K} R_n(\Pi, \nu)$$

as the **minimax regret** since it is minimum worst-case regret across policies and arm distributions. Note that the minimax regret is dependent solely on the family of the MABs.

We can now begin our analysis with the definitions out of the way. We consider a MAB family of unit-variance Gaussian arms -

$$\mathcal{E} = \mathcal{E}_{\mathcal{N}}(1)$$

Since all the arm distributions have a unit variance, we can simply use their means to characterize them. Thus, $\mathcal{E}_{\mathcal{N}}(1)$ can be parameterized by $\mu = (\mu_1, \dots, \mu_K)$.

Theorem 4.3. Let $K > 1$ and $n > K - 1$. Then for any policy Π , $\exists$ an instance $\mu \in \mathcal{E}_{\mathcal{N}}(1)$ where $\mu_i \in [0, 1]$ for all arms i such that -

$$R_n(\Pi, \mu) \geq \frac{1}{27} \sqrt{n(K-1)}$$

We see that we are taking a very specific subset of families of $\mathcal{E}$. But if the supremum of the regret is higher than some value, then that will also apply for the larger family as well.

Proof. Fix some policy Π and then come up with an instance -

$$\mu = [\Delta, 0, 0, \dots, 0]$$

where $\Delta \in (0, \frac{1}{2})$. Thus, it can be seen that arm 1 is the optimal arm. Now consider the arm i such that -

$$i = \arg \min_{i \neq j} \mathbb{E}_{\mu}[T_i(n)]$$

Which is the arm that my policy Π pulls the least amount of times. Now, let us construct another instance -

$$\mu' = [\Delta, 0, \dots, 0, 2\Delta, 0, \dots, 0]$$

Where now the arm i has a mean of 2Δ which makes it optimal. Note that μ, μ' both induce two different probability measures $\mathbb{P}_{\mu}, \mathbb{P}_{\mu'}$. Now, consider the event -

$$A = \left\{ T_1(n) \leq \frac{n}{2} \right\}$$

For μ , arm 1 is optimal which means that event A is bad for $\mathbb{P}_{\mu}$. Additionally, A^C is a bad event of μ' since arm 1 is sub-optimal and therefore pulling it more than half times is a bad event for $\mathbb{P}_{\mu'}$. One can see that we are planning to use the BH inequality at some point.

$$R_n(\Pi, \mu) \geq \mathbb{P}_{\mu}(A) \cdot \frac{n\Delta}{2} \quad ; \quad R_n(\Pi, \mu') \geq \mathbb{P}_{\mu'}(A^C) \cdot \frac{n\Delta}{2}$$

Then, we can write -

$$\begin{aligned} R_n(\Pi, \mu) + R_n(\Pi, \mu') &\geq \frac{n\Delta}{2} (\mathbb{P}_{\mu}(A) + \mathbb{P}_{\mu'}(A^C)) \\ &\geq \frac{n\Delta}{4} \exp \{ -\text{KL}(\mathbb{P}_{\mu}, \mathbb{P}_{\mu'}) \} \end{aligned}$$

Which follows from BH inequality. From divergence-decomposition lemma, we get -

$$\text{KL}(\mathbb{P}_{\mu}, \mathbb{P}_{\mu'}) = \sum_{i=1}^K \mathbb{E}_{\mu}[T_i(n)] \cdot \text{KL}(\mu, \mu')$$

Substituting back, we get -

$$R_n(\Pi, \mu) + R_n(\Pi, \mu') \geq \frac{n\Delta}{4} \exp \left\{ -\mathbb{E}_{\mu}[T_i(n)] \cdot \frac{(2\Delta)^2}{2} \right\}$$

Since i is a sub-optimal arm under μ , we get -

$$\mathbb{E}_{\mu}[T_i(n)] \leq \frac{n}{K-1}$$

Substituting back, we get -

$$R_n(\Pi, \mu) + R_n(\Pi, \mu') \geq \frac{n\Delta}{4} \exp \left\{ \frac{-2n\Delta^2}{K-1} \right\}$$

Optimizing (maximizing) over Δ , we get -

$$\Delta = \sqrt{\frac{K-1}{4n}} \implies R_n(\Pi, \mu) + R_n(\Pi, \mu') \geq \frac{\sqrt{n(K-1)}}{8\sqrt{e}}$$

Thus, we get -

$$\max \{R_n(\Pi, \mu), R_n(\Pi, \mu')\} \geq \frac{\sqrt{n(K-1)}}{16\sqrt{e}} \geq \frac{\sqrt{n(K-1)}}{27}$$

Hence, we get -

$$\boxed{R_n(\Pi, \mu) \geq \frac{1}{27}\sqrt{n(K-1)}}$$

□

For the instance-independent UCB we had discussed previously, the regret bound was of the order $O(\sqrt{nK \ln(n)})$. We can see that theoretic lower bound of the minimax regret is $O(\sqrt{nK})$. Thus, the instance-independent UCB isn't too bad.

Consistent policy

A policy Π is said to be consistent on a MAB family $\mathcal{E}$ if for all $\nu \in \mathcal{E}^K$, the regret $R_n(\Pi, \nu) = o(n^a)$ i.e. the regret grows slower than any polynomial.

We can see that UCB is consistent since it has a logarithmic regret.

Let $\mu^* \in \mathbb{R}, \nu_i \in \mathcal{E}$ such that $\mu(\nu_i) \leq \mu^*$. Basically, we have a family of distribution and an instance such that there exists some arm with mean reward less than a constant μ^* . Now, we can define -

$$d_{\inf}(\nu_i, \mu^*, \mathcal{E}) = \inf \{KL(\nu_i, \nu'_i) : \nu'_i \in \mathcal{E}, \mu(\nu'_i) > \mu^*\}$$

Basically, the above quantity is the smallest KL-divergence perturbation to be made to ν_i such that the mean of the distribution goes beyond μ^* . Another way to think about this is that $d_{\inf}(\nu_i, \mu^*, \mathcal{E})$ is the smallest perturbation in KL sense needed to make a sub-optimal arm optimal. Note that, the perturbation should not be outside the family $\mathcal{E}$.

With this definition done, we can come up with an instance dependent lower bound.

Theorem 4.4. Let Π be a consistent policy over $\mathcal{E}^K$. Then, for any $\nu \in \mathcal{E}^K$, we get -

$$\liminf_{n \rightarrow \infty} \frac{R_n(\Pi, \nu)}{\ln(n)} \geq \sum_{i: \Delta_i > 0} \frac{\Delta_i}{d_{\inf}(\nu_i, \mu^*, \mathcal{E})}$$

Proof. Consider the notation $d_i = d_{\inf}(\nu_i, \mu^*, \mathcal{E})$ for brevity. Then, we shall aim to prove -

$$\mathbb{E}_\nu[T_i(n)] \geq \frac{1}{d_i}$$

As in, a higher d_i implies that the arm is far from the optimal arm which means that it should be pulled lesser in expectation.

Consider a sub-optimal arm i . Then, there exists a perturbed version ν'_i such that -

- $\mu(\nu'_i) > \mu^*$
- $\text{KL}(\nu_i, \nu'_i) \leq d_i + \varepsilon$

The second condition follows since d_i is the infimum over all perturbations that ensure the mean of the sub-optimal arm is greater than the mean of the optimal arm. Thus, we know that ν'_i exists and we can define a new instances where ν_i is replaced by ν'_i and the rest are the same. From divergence-decomposition lemma, we get -

$$\text{KL}(\mathbb{P}_\nu, \mathbb{P}_{\nu'}) = \sum_{i=1}^K \mathbb{E}_\nu[T_i(n)] \cdot \text{KL}(\nu, \nu') \leq \mathbb{E}_\nu[T_i(n)] \cdot (d_i + \varepsilon)$$

This is because the other arms are unchanged so the KL divergences will be 0. Now, define an event -

$$A = \left\{ T_i(n) > \frac{n}{2} \right\}$$

We can see that since i is a sub-optimal arm, event A is a bad event in the original instance. In the perturbed instance, A^C is also a bad event. We can say that -

$$\begin{aligned} R_n(\pi, \nu) &\geq \mathbb{P}_\nu(A) \cdot \frac{n\Delta_i}{2} \\ R_n(\pi, \nu') &\geq \mathbb{P}'_\nu(A^C) \cdot \frac{n(\mu(\nu'_i) - \mu^*)}{2} \end{aligned}$$

Thus, we get -

$$R_n(\pi, \nu) + R_n(\pi, \nu') \geq \frac{n}{2} \min[\Delta_i, (\mu(\nu'_i) - \mu^*)] \left\{ \mathbb{P}_\nu(A) + \mathbb{P}'_\nu(A^C) \right\}$$

Let us denote $c = \min[\Delta_i, (\mu(\nu'_i) - \mu^*)]$. Then, using BH-inequality, we get -

$$\begin{aligned} R_n(\pi, \nu) + R_n(\pi, \nu') &\geq \frac{nc}{4} \exp\{-\text{KL}(\mathbb{P}_\nu, \mathbb{P}'_\nu)\} \\ &\geq \frac{nc}{4} \exp\{-\mathbb{E}_\nu[T_i(n)] \cdot (d_i + \varepsilon)\} \end{aligned}$$

Taking log on both sides, we get -

$$\ln[R_n(\pi, \nu) + R_n(\pi, \nu')] \geq \ln\left(\frac{c}{4}\right) + \ln(n) - \mathbb{E}_\nu[T_i(n)] \cdot (d_i + \varepsilon)$$

Since the policy Π is consistent (from assumption), then we can say that -

$$\ln[R_n(\pi, \nu) + R_n(\pi, \nu')] = o(\ln(n))$$

Why is this true? Left to prove by the reader. Dividing both sides by $\ln(n)$ and taking the limit, we get -

$$\begin{aligned} \liminf_{n \rightarrow \infty} \frac{R_n(\pi, \nu) + R_n(\pi, \nu')}{\ln(n)} &\geq \liminf_{n \rightarrow \infty} \left\{ \frac{\ln\left(\frac{c}{4}\right)}{\ln(n)} + 1 - \frac{\mathbb{E}_\nu[T_i(n)] \cdot (d_i + \varepsilon)}{\ln(n)} \right\} \\ \liminf_{n \rightarrow \infty} \frac{R_n(\pi, \nu) + R_n(\pi, \nu')}{\ln(n)} &\geq \liminf_{n \rightarrow \infty} \left\{ 1 - \frac{\mathbb{E}_\nu[T_i(n)] \cdot (d_i + \varepsilon)}{\ln(n)} \right\} \end{aligned}$$

Complete the proof. □

Examples of distribution families where $d_i > 0$ for all sub-optimal arms i

- σ -subGaussians
- $\mathcal{E}_{\alpha,\beta} = \{\nu: \mathbb{E}[|\nu|^\alpha] \leq \beta\}$

For an example, consider the family of arms that are sampled from a unit-variance Gaussians. Then -

$$d_i = \frac{\Delta_i^2}{2}$$

Thus, we get -

$$\liminf_{n \rightarrow \infty} \frac{R_n}{\ln(n)} \geq \sum_{i: \Delta_i > 0} \frac{2}{\Delta_i}$$

In the first UCB algorithm, the regret was upper bounded as -

$$R_n \leq 3 \sum_{i: \Delta_i > 0} \Delta_i + \sum_{i: \Delta_i > 0} \frac{16 \ln(n)}{\Delta_i}$$

Thus, we can see that the UCB algorithm is optimal (compared to the theoretical lower bound) up to some constant, which is really good.

Note

We can have families where $d_i = 0$ which means that we can increase the mean by a large margin by just changing the KL divergence by an infinitesimally small amount. For example -

$$\mathcal{E} = \left\{ \bigcup_{\sigma > 0} \sigma - \text{subGaussian} \right\}$$

For such a class, getting logarithmic regret is **impossible**. For such classes, algorithms like UCB cannot be consistent. More information can be found here.

Pure Exploration bandits - Fixed budget

This is essentially the **best-arm identification** problem where the instance explores all the arms and aims to identify the best-arm. However, the caveat here is that there is a **fixed budget** which is the maximum number of arm pulls (overall) you can perform a.k.a the horizon. The metric used here is the confidence/error probability of identifying the best-arm.

Consider an instance with K arms with ν_i being the reward distribution of arm i , with μ_i being the mean reward of arm i . WLOG, assume that $\mu_1 \geq \mu_2 \geq \dots \geq \mu_K$. Additionally, we are given a budget of n pulls and after these n pulls, the algorithm is supposed to return the estimate of the best arm $\hat{a}$. The metric will be the error probability -

$$p_e = \mathbb{P}\{\hat{a} \neq a^*\}$$

Algorithm 11 Uniform Exploration

Require: Number of arms K , budget n , distribution $\nu = \{\nu_i\}_{i=1}^n$

Pull each arm $\frac{n}{K}$ times

Calculate $\hat{a} = \arg \max_i \hat{\mu}_i$

In the mid-sem, this question was asked and since we were dealing with 1-subG arms, the analysis followed cleanly since probability calculations are trivial. However, let us now check a more general case. We write -

$$\begin{aligned}
p_e^{UE} &= \sum_{i=2}^K \mathbb{P}(\hat{\mu}_i \geq \hat{\mu}_1) \\
&= \sum_{i=2}^K \mathbb{P}\{(\hat{\mu}_1 - \mu_1) + (\mu_i - \hat{\mu}_i) \leq -\Delta_i\} \\
&= \sum_{i=2}^K \left[\mathbb{P}(\hat{\mu}_1 - \mu_1 \leq \frac{-\Delta_i}{2}) \cup \mathbb{P}(\mu_i - \hat{\mu}_i \leq \frac{-\Delta_i}{2}) \right] \\
&\leq \sum_{i=2}^K \left[\mathbb{P}(\hat{\mu}_1 - \mu_1 \leq \frac{-\Delta_i}{2}) + \mathbb{P}(\mu_i - \hat{\mu}_i \leq \frac{-\Delta_i}{2}) \right] \\
&\leq 2 \sum_{i=2}^K \exp \left\{ -\frac{\Delta_i^2 n}{8K} \right\}
\end{aligned}$$

The last expression follows when we assume the arms are 1-subG (just for a final expression). It is clear to see that for a higher budget, the algorithm error bound decays exponentially. Additionally, if the arms are well-separated i.e. Δ_i are high for non-optimal arms, the error decay rate is higher and that makes sense. Thus, the actual decay rate for 11 will be $\frac{\Delta_2^2}{8K}$.

$$p_e^{UE} \leq 2 \sum_{i=2}^K \exp \left\{ -\frac{\Delta_i^2 n}{8K} \right\} \leq 2K \exp \left\{ -\frac{\Delta_2^2 n}{8K} \right\}$$

If the closest sub-optimal arm is fairly well-separated, then the error decay will be faster.

Successive Rejects

One main problem with 11 is that whether an arm is close to optimal or is highly sub-optimal, they both get the same amount of exploration. If we already know that an arm is highly sub-optimal, why would we want to explore them again?

Algorithm 12 Successive Rejects Exploration

Require: Number of arms K , budget n , distribution $\nu = \{\nu_i\}_{i=1}^n$
Consider $n_i \in \mathbb{N}$ where $i \in [K-1]$ and $n_i \leq n_{i+1}$ such that

$$\sum_{i=1}^{K-2} n_i + 2n_{K-1} \leq n$$

for $i = 1$ to $K-1$ **do**

The $i^{[th]}$ phase has started.

Give each arm $n_i - n_{i-1}$ pulls.

Eliminate arm $j = \arg \min_i \hat{\mu}_i(n_i)$

end for

The algorithm in 12 can be visualized as a tournament that happens in **phases** where each phase is indexed by i . The value n_i denotes the number of pulls each arm gets in phase i . In each phase, the worst-performing arm is rejected and the other arms move on. So, in phase i , there are $K - n + i$ surviving arms. After $K-1$ phases, we are left with an estimate of the best arm. Now, notice the condition -

$$\sum_{i=1}^{K-2} n_i + 2n_{K-1} = n_1 + n_2 + n_3 + \dots + n_{K-2} + 2n_{K-1} \leq n$$

Let us break this down. In a phase $i \in [K-2]$, only the arm that gets kicked out at phase i gets exactly n_i pulls. However, in $K-1$ phase, there are only two arms left, each of whom gets n_{K-1} pulls in total and that is why we add $2n_{K-1}$. All of these has to be less than n .

Note

One important distinction needs to be noted. n_2 denotes the total number of pulls each arm gets by the end of phase 2. That means that in phase 2, each arm that survived phase 1 gets an additional $n_2 - n_1$ pulls. So if we say $n_1 = n_2$, this means that in phase 2, no additional pulls are made to any arm and another arm is rejected. Thus, $n_1 = n_2$ basically implies that two arms are being kicked out after the same number of pulls. Thus, if we write -

$$n_1 = n_2 = \dots = n_{K-1} = \frac{n}{K}$$

this corresponds to 11.

Now, let us analyse the error probability. Consider an error event E_i which is the event that arm 1 (optimal) gets kicked out in phase i . Then, we can write -

$$p_e^{SR} = \sum_{i=1}^{K-1} \mathbb{P}(E_i)$$

Analysis Intuition

In phase i , at least one of the worst i arms has participated (since only $i - 1$ eliminations have taken place so far) and if arm 1 gets kicked out, then it lost to at least one of the worst i performing arms. Thus, we can write -

$$\mathbb{P}(E_i) \leq \mathbb{P} \left(\bigcup_{j=K-i+1}^K \hat{\mu}_1(n_i) \leq \hat{\mu}_j(n_i) \right)$$

Using the intuition, we get -

$$\begin{aligned} p_e^{SR} &= \sum_{i=1}^{K-1} \mathbb{P}(E_i) \leq \sum_{i=1}^{K-1} \mathbb{P} \left(\bigcup_{j=K-i+1}^K \hat{\mu}_1(n_i) \leq \hat{\mu}_j(n_i) \right) \\ &\leq \sum_{i=1}^{K-1} \sum_{j=K-i+1}^K \mathbb{P}(\hat{\mu}_1(n_i) \leq \hat{\mu}_j(n_i)) \end{aligned}$$

Using the previous analysis for 1-subG arms, we get -

$$\sum_{j=K-i+1}^K \mathbb{P}(\hat{\mu}_1(n_i) \leq \hat{\mu}_j(n_i)) \leq \sum_{j=K-i+1}^K 2 \exp \left\{ -\frac{n_i \Delta_j^2}{8} \right\} \leq 2i \exp \left\{ -\frac{n_i \Delta_{K-i+1}^2}{8} \right\}$$

Thus, we get -

$$p_e^{SR} \leq \sum_{i=1}^{K-1} 2i \exp \left\{ -\frac{n_i \Delta_{K-i+1}^2}{8} \right\}$$

Now, optimizing over n_i shall indeed give the tightest bounds. The actual SR algorithm using the following elimination schedule -

$$n_i = \left\lceil \frac{n - K}{\log(K)(K - i + 1)} \right\rceil \quad \text{where} \quad \log(K) = \frac{1}{2} + \sum_{i=2}^K \frac{1}{i}$$

Effectively, $n_i \propto \frac{1}{K-i+1}$ that is the total number of pulls in the i^{th} phase should get pulls that are inversely proportional to the number of sub-optimal arms left in the tournament. If we use this elimination schedule, then our error probability becomes -

$$p_e^{SR} \leq \sum_{i=1}^{K-1} 2i \exp \left\{ -\frac{(n - K) \Delta_{K-i+1}^2}{8 (\log(K)(K - i + 1))} \right\}$$

Consider the arm j such that -

$$j = \arg \min_i \frac{\Delta_{K-i+1}^2}{K-i+1}$$

Subsequently, we define -

$$H_2 = \max_{i \geq 2} \frac{i}{\Delta_i^2}$$

Using these values, we can get the final bound as -

$$p_e^{SR} \leq 2K^2 \exp \left\{ -\frac{(n-K)}{8\log(K)H_2} \right\}$$

Note

Consider the decay rates of 11 and 12.

$$\alpha^{UE} \geq \frac{\Delta_2^2}{8K}$$

$$\alpha^{SR} \geq \frac{1}{8\log(K)H_2}$$

Pure Exploration bandits - Fixed Confidence

Here, we are given a certain confidence level and our algorithm needs to be right with that confidence. Given this, the algorithm decides to return the stopping time/number of samples needed to get that confidence, which becomes our metric. There are three blocks - sampling algorithm, the stopping rule and then the optimal arm that the algorithm returns.

δ -probably correct

An algorithm is said to be δ -PC if -

$$\mathbb{P}(\tau < \infty, \hat{a} \neq 1) \leq \delta$$

That is, the probability of the algorithm stops sampling and returns a sub-optimal arm is upper bounded by δ . We are assuming that arm 1 is optimal.

Interestingly, suppose my algorithm doesn't stop at all? Then the probability of stopping and returning the sub-optimal arm is 0 which will be less than δ and hence is also δ -PC. So this is a low bar to meet. So the task now is to find an algorithm that is δ -PC that minimizes $\mathbb{E}[\tau]$ - expected stopping time.

Theorem 4.5. Consider a 1-Gaussian instance (each arm is a Gaussian with variance 1) with means $\mu = \{\mu_1, \dots, \mu_K\}$ such that $\mu_1 \geq \mu_2 \geq \dots \geq \mu_K$ (arm 1 is optimal). Then, for any δ -PC algorithm,

$$\mathbb{E}[\tau] \geq 2 \ln \left(\frac{1}{4\delta} \right) \sum_{i=1}^K \frac{1}{\Delta_i^2}$$

where $\Delta_1 = \Delta_2 = \mu_1 - \mu_2$.

The expected stopping time goes up as δ decreases since we are now expected to be more sure. Also, if the sub-optimality gaps are smaller, identifying the optimal arm is harder, thus increasing the expected stopping time.

Note

We can see that if the optimal arm is not unique, then $\mathbb{E}[\tau] \rightarrow \infty$. This comes from the fact that we are unable to check which arm is optimal, even though both are optimal. Thus, unlike the case of fixed budget, we **NEED** the optimal arm to be unique.

Proof. We start off by creating a new instance of each arm i . These new instances are all 1-Gaussian. For the optimal arm 1, we can define a new instance as -

$$\mu_i^{[1]} = \begin{cases} \mu_2 - \epsilon & i = 1 \\ \mu_i & i \neq 1 \end{cases}$$

Basically, we are keeping all the arms same but the optimal arm is now μ_2 with arm 1 being the second best. For any other arm, we define new instances as -

$$\mu_i^{[a]} = \begin{cases} \mu_a + \Delta_a + \epsilon & i = a \\ \mu_i & i \neq a \end{cases}$$

Basically, arm a becomes optimal. This unique choice of method will make sense soon. Using BH-inequality and Divergence-decomposition lemma, we can write -

$$\mathbb{P}_\mu(A^C) + \mathbb{P}_{\mu^{[a]}}(A) \geq \frac{1}{2} \exp \left\{ -\mathbb{E}_\mu[N_a(\tau)] \cdot \text{KL}(\mu, \mu^{[a]}) \right\}$$

where A is the event that after some finite stopping time we return the arm that is not optimal in the modified instance.

$$A = \left\{ \tau < \infty \quad , \quad \hat{a} \neq a_{\mu^{[a]}}^* \right\}$$

Since the algorithm is δ -PC, we can write -

$$\mathbb{P}_{\mu^{[a]}}(A) \leq \delta$$

Also, we can write -

$$A^C = \{ \tau = \infty \} \cup \left\{ \tau < \infty \quad , \quad \hat{a} = a_{\mu^{[a]}}^* \right\}$$

Again using δ -PC, we get -

$$\mathbb{P}(A^C) \leq \mathbb{P}(\{ \tau = \infty \}) + \mathbb{P}(\{ \tau < \infty, \hat{a} = a_{\mu^{[a]}}^* \}) \leq \delta$$

Thus, we get -

$$4\delta \geq \exp \left\{ -\mathbb{E}_\mu[N_a(\tau)] \cdot \text{KL}(\mu, \mu^{[a]}) \right\}$$

Since we are assume 1-Gaussian arms and the perturbation done is by some ϵ quantum, we can write -

$$\text{KL}(\mu, \mu^{[a]}) = \frac{(\Delta_a + \epsilon)^2}{2}$$

Thus, we get -

$$4\delta \geq \exp \left\{ -\mathbb{E}_\mu[N_a(\tau)] \cdot \frac{(\Delta_a + \epsilon)^2}{2} \right\}$$

Re-arranging and using the fact that $\epsilon \approx 0$ gives us -

$$\mathbb{E}_\mu[\tau] \geq 2 \ln \left(\frac{1}{4\delta} \right) \sum_{a=1}^K \frac{1}{\Delta_a^2}$$

□

Note

The divergence-decomposition lemma holds only when we know the horizon i.e. the total number of pulls we make. However, in the derivation above, we have used the lemma for expected stopping times as well, which is valid. Check [1] for more information.

Additionally, we have assumed that $\mathbb{E}_\mu[\tau] < \infty$. If that is not the case, the lower bound follows anyways lol.

Algorithm 13 Action Elimination

Require: Number of arms K , confidence δ , distribution $\mu = \{\mu_i\}_{i=1}^n$

Initialize: Active set $A = [K]$

for $t < \infty$ **do**

 Pull each arm in A once

for Arms $a = 1$ to K **do**

 Calculate $\text{UCB}_a(t) = \hat{\mu}_a^t + \sqrt{\frac{2}{t} \ln \left(\frac{2Kt^2C}{\delta} \right)}$

 Calculate $\text{LCB}_a(t) = \hat{\mu}_a^t - \sqrt{\frac{2}{t} \ln \left(\frac{2Kt^2C}{\delta} \right)}$

end for

 Elimination set $E = \{i \in A : \exists j \in A \text{ s.t. } \text{UCB}_i(t) \leq \text{LCB}_j(t)\}$

$A \leftarrow A \setminus E$

if $|A| = 1$ **then**

return $a \in A$

STOP

else

$t = t + 1$

end if

end for

The algorithm in 13 samples each arm in a round robin fashion. At each time step, if there is an arm that has a UCB lower than the LCB of any other arm, then that arm is sub-optimal and therefore is eliminated. The sole remaining arm after all others have been eliminated is the optimal arm and is returned.

Claim 4.6. *The algorithm as defined in 13 is δ -PC*

Proof. Let us define a “bad event” for an arm i after t pulls as -

$$B_i(t) = \{|\hat{\mu}_i(t) - \mu_i| > \alpha_t\}$$

where

$$\alpha_t = \sqrt{\frac{2}{t} \ln \left(\frac{2Kt^2C}{\delta} \right)}$$

This event corresponds to the fact that the UCB and LCB are invalid for arm i after t pulls. Using subG bounds, we write -

$$\mathbb{P}(B_i(t)) \leq 2 \exp \left\{ -\frac{t\alpha_t^2}{2} \right\} = \frac{\delta}{Kt^2c}$$

Taking union over all arms and time steps, we get -

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}\left[\bigcup_{i \in [K]} \bigcup_{t=1}^{\infty} B_i(t)\right] \\ &\leq \sum_{i \in [K]} \sum_{t=1}^{\infty} \mathbb{P}(B_i(t)) = \frac{\delta}{c} \sum_{t=1}^{\infty} \frac{1}{t^2} \\ &\leq \delta \quad \text{where} \quad c \geq \frac{\pi^2}{6}\end{aligned}$$

Hence, the algorithm is δ -PC □

Now, B^C is the event that 13 stops in finite time and returns the optimal arm. Consider a sub-optimal arm $i \neq 1$ and the optimal arm 1. When do we eliminate arm i ? That happens when -

$$\max \text{UCB}_i(t) < \min \text{LCB}_1(t)$$

We know that for any arm i , we have

$$\text{UCB}_i(t) - \text{LCB}_i(t) = 2\alpha_t$$

Thus, we know that if $\alpha_t \leq \frac{\Delta_i}{4}$, then no matter what happens, $\max \text{UCB}_i(t) < \min \text{LCB}_1(t)$. Thus, for each sub-optimal arm $i \neq 1$, we see that the number of pulls they get is given by -

$$T_i = \min_t \alpha_t \leq \frac{\Delta_i}{4}$$

Thus, the total number of pulls across all arms becomes -

$$\tau \leq \sum_{i=1}^K T_i = O\left(\frac{1}{\Delta_i^2} \ln \left\lceil \frac{K}{\delta \Delta_i} \right\rceil\right)$$

Problem

Consider the case where we have more than one sub-optimal arms that have the same mean/distribution. Then, if the optimal arm gets eliminated previously, there can be a case where the confidence intervals of these arms will always overlap, which prevents them from being eliminated and therefore the algorithm 13 doesn't stop.

LUCB

This algorithm aims to fix the problem as mentioned above for 13. It basically compares the top-two arms instead of all arms.

Defining the “top-two arms” is done in a specific way - the top arm is the arm with the highest empirical mean while the competitor (second-top) arm is the arm (apart from the top arm) that has the highest UCB. Note that in this algorithm (as mentioned in 14), there is **no arm elimination**.

Also note that since each arm gets different number of pulls, the α_t value is now arm dependent -

$$\alpha_{i,t} = \sqrt{\frac{1}{2N_i(t-1)} \ln \left(\frac{cKt^4}{\delta} \right)}$$

Algorithm 14 LUCB

Require: Number of arms K , confidence δ , distribution $\mu = \{\mu_i\}_{i=1}^n$
Initialize: Active set $A = [K]$
for $t < \infty$ **do**
 Pull each arm in A once
 Define: The top arm $a_{\text{top}} = \arg \max_{i \in [K]} \hat{\mu}_i(t)$
 Define: The competitor arm $a_{\text{comp}} = \arg \max_{i \in [K], i \neq a_{\text{top}}} \hat{\mu}_i(t) + \alpha_{i,t}$
 Pull a_{top} and a_{comp} once.
 if $\text{LCB}_{a_{\text{top}}} > \text{UCB}_{a_{\text{comp}}}$ **then**
 return Optimal arm a_{top}
 STOP
end if
end for

Claim 4.7. • The algorithm 14 stops with probability 1. There is no scenario where it doesn't stop.
 • The algorithm 14 is δ -PC.

Proof. **TO BE FILLED LATER**

□

Claim 4.8. Consider a variation of 14 where

$$\alpha_{i,t} = \sqrt{\frac{2}{N_i(t-1)} \ln \left(\frac{2KN_i^2(t-1)c}{\delta} \right)}$$

Then, this algorithm is δ -PC (to be proven by reader). Now suppose B is an event where at least one confidence intervals for at least one arm at some time step misbehaves. Then, on B^C , i.e. where all the confidence intervals for all arms across all time steps are valid, if the algorithm 14 hasn't stopped yet implies that either a_{top} or a_{comp} is bad.

Proof. Consider the top-two arms—1, 2—and we get $c = \frac{\mu_1 + \mu_2}{2}$. Now, arm 1 is considered to be **BAD** if $\hat{\mu}_1 - \alpha_{1,t} < c$. For any other arm $i \neq 1$, arm i is considered to be **BAD** if $\hat{\mu}_i + \alpha_{i,t} > c$. Basicall, if the LCB of the optimal arm drops below c , it is bad. Similarly, if the UCB of the sub-optimal arm is above c , then it is considered to be bad. We define B similar to the previous case where at least one confidence interval misbehaves. □

$$\tau \leq \sum_{t=1}^{\infty} \sum_{i=1}^K \mathbb{1} \{ (a_{\text{top}} = i \text{ OR } a_{\text{comp}} = i) \text{ AND arm } i \text{ is BAD} \} = \sum_{i=1}^K \tilde{T}_i$$

Basically, if the algorithm hasn't stopped yet, then either one of the top-2 arms are BAD as mentioned in the claim before, which helps us define the total number of pulls before stopping 14 performs.

Track & Stop

5 Markov Decision Processes (MPDs)

The general process in most RL applications is as follows -

1. The agent observes the environment state S_t .
2. An agent performs an action A_t given S_t .
3. The environment state evolves to S_{t+1} based on S_t, A_t
4. The environment returns a reward (R_t)/cost (C_t) based on S_t, A_t

The aim of the agent is to minimize cost (or maximize reward).

Markov Chain

A random process $\{X_n\}_{n \geq 0}$ taking values in countable set S is a *discrete-time Markov Chain* if $\forall i_0, i_1, \dots, i_{n-1}, i, j \in S$ we have -

$$\mathbb{P}(X_{n+1} = j \mid X_0 = i_0, X_1 = i_1, \dots, X_n = i) = \mathbb{P}(X_{n+1} = j \mid X_n = i)$$

In simpler terms, the state at X_{n+1} is independent of $X_0, \dots, X_{n-1}$ given we know X_n . This is also called the **Markovian property**. If the probability values of state transitions (also called **transition probability**) are static with time i.e. the probability of going from state i to j doesn't change with time, then the MC is said to be **time homogeneous**.

Controlled Markov Property

The probability of transitioning to a state X_{n+1} at time $n+1$ is independent of not just the previous states, but also the previous actions and costs given we know X_n and the action taken at time n .

$$\begin{aligned} \mathbb{P}(X_{n+1} = j \mid X_0 = i_0, A_0 = a_0, C_0, c_0, X_1 = i_1, A_1 = a_1, C_1, c_1, \dots, X_n = i, A_n = a_n) \\ = \mathbb{P}(X_{n+1} = j \mid X_n = i, A_n = a_n) \end{aligned}$$

Now we can formally define a **MDP** as follows -

$$\text{MDP} = M(S, A, P, C, \alpha)$$

where -

- S - A countable state space.
- A - A finite action space.
- P - The transition matrix that contains the probability of transitioning from one state to another.
- C - The cost matrix that stores the cost for state-action pairs.
- α - The discounting factor belonging to $(0, 1)$.

Note

In an asymptotic sense, there are two ways of defining the overall cost. One way to do this is by taking the expectation of the total cost. On the other hand, we can add a *discounting factor* that models the fact that the **weight** of the cost at time t goes down as the time goes on.

Discounted-cost, Infinite-horizon MDPs

Let us define a **policy** (π) which gives the action to be taken given the current state.

Value Function

For a policy $\pi \in \Pi$, the value function of π represented as V_π is the expected infinite-horizon, discounted cost given the MDP starts at some state $i \in S$.

$$V_\pi(i) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \alpha^t C(X_t, A_t) \mid X_0 = i \right]$$

A compilation of these value functions across state can be written as -

$$V_\pi = [V_\pi(i) \forall i \in S]$$

Thus, the value function denotes the expected cost/reward the policy will incur/earn asymptotically, given it starts at some state $i \in S$. The optimal value function for state $i \in S$ is the value function of the optimal policy π^* at state i and is defined as -

$$V^*(i) = \inf_{\pi} V_\pi(i)$$

Naturally, a policy $\pi^* \in \Pi$ is said to be an optimal policy if $V_\pi(i) = V^*(i)$ for all states $i \in S$. Note that a single MDP can have more than one optimal policies, but the optimal value function is unique.

Theorem 5.1. Consider an MDP $M = (S, A, T, R, \gamma)$ and a set of policies Π such that $\forall \pi \in \Pi$, π is -

- **Markovian** - The policy action depends solely on the previous state.
- **Deterministic** - The action taken at a state by the policy is with probability 1.
- **Stationary** - The policy doesn't update with time

Then, M is **guaranteed** to have an optimal policy $\pi^* \in \Pi$ such that $V^*(s) \geq V_\pi(s)$ for all $s \in S$ and $\pi \in \Pi$

The proof of this theorem comes later on. This goes to show given an MDP with policies that are Markovian, deterministic and stationary, there will be an optimal policy that gives the optimal value function.

Bellman equations

One of the ways to quantify how good our policy π is to check its value function for each state. But we can't run the MDP asymptotically starting at each state just to get the value function. This is where the **Bellman equations** come in.

Theorem 5.2. Consider a discounted MDP $M(S, A, P, C, \alpha)$. Then, we can write -

$$V_\pi(s) = C(s, \pi(s), s') + \sum_{s' \in S} \alpha P(s, \pi(s), s') \cdot V_\pi(s')$$

Proof. For the policy and MDP defined above, we can write -

$$\begin{aligned} V_\pi(s) &= \mathbb{E}_\pi [c_0 + \alpha c_1 + \alpha^2 c_2 + \dots \mid s_0 = s] \\ &= \sum_{s' \in S} P(s, \pi(s), s') \mathbb{E}_\pi [c_0 + \alpha c_1 + \alpha^2 c_2 + \dots \mid s_0 = s, s_1 = s'] \end{aligned}$$

Now, we can use standard probability identities to write the following -

$$\begin{aligned}
V_\pi(s) &= \sum_{s' \in S} P(s, \pi(s), s') \mathbb{E}_\pi [c_0 + \alpha c_1 + \alpha^2 c_2 + \alpha^3 c_3 + \dots | s_0 = s, s_1 = s'] \\
&= \sum_{s' \in S} P(s, \pi(s), s') \mathbb{E}_\pi [r_0 | s_0 = s, s_1 = s'] + \alpha \sum_{s' \in S} P(s, \pi(s), s') \mathbb{E}_\pi [c_1 + \alpha c_2 + \alpha^2 c_3 + \dots | s_0 = s, s_1 = s'] \\
&= X + Y \quad (\text{Say})
\end{aligned}$$

Let us consider the case of X . We know from definition that $\mathbb{E}_\pi [c_0 | s_0 = s, s_1 = s']$ is the expected cost that environment gives when the start state is s , the next state is s' given the action is $\pi(s)$.

$$\mathbb{E}_\pi [c_0 | s_0 = s, s_1 = s'] = C(s, \pi(s), s')$$

Now, let us consider Y . The expectation term is the expected sum of rewards (with discount) when the start state is s' . Since the reward c_0 is not considered, the state $s_0 = s$ makes no difference to the term. Thus, we can write -

$$\mathbb{E}_\pi [c_1 + \alpha c_2 + \alpha^2 c_3 + \dots | s_0 = s, s_1 = s'] = V_\pi(s')$$

Substituting back, we get -

$$\begin{aligned}
V_\pi(s) &= \sum_{s' \in S} P(s, \pi(s), s') C(s, \pi(s), s') + \alpha \sum_{s' \in S} P(s, \pi(s), s') V_\pi(s') \\
&= \sum_{s' \in S} P(s, \pi(s), s') [C(s, \pi(s), s') + \alpha V_\pi(s')]
\end{aligned}$$

However, we know that the cost is for a particular state and action i.e. $C(s, \pi(s), s')$ is the same irrespective of the s' . Thus, we get -

$$\begin{aligned}
V_\pi(s) &= \sum_{s' \in S} P(s, \pi(s), s') \cdot C(s, \pi(s), s') + \alpha \sum_{s' \in S} P(s, \pi(s), s') \cdot V_\pi(s') \\
&= C(s, \pi(s), s') \sum_{s' \in S} P(s, \pi(s), s') + \alpha \sum_{s' \in S} P(s, \pi(s), s') \cdot V_\pi(s') \\
&= C(s, \pi(s), s') + \alpha \sum_{s' \in S} P(s, \pi(s), s') \cdot V_\pi(s')
\end{aligned}$$

□

These are the **Bellman equations**, where there will be one equation per state and each equation uses the value function of every other state. Hence, for S states, we have S linear equations and S unknowns. Finding the value function is now just the process of solving a system of linear equations which is simple and easy.

Lemma 5.3. Consider a discounted MDP $M(S, A, P, C, \alpha)$. Then, we can write -

$$V^*(i) = \min_{a \in A} \left[c_{i,a} + \alpha \sum_{j \in S} p_{i,j}(a) V^*(j) \right] \quad \forall i \in S$$

Proof. Consider an arbitrary policy $\pi \in \Pi$. Then,

$$V_\pi(i) = \sum_{a \in A} P_\pi(a) \left[c(i, a) + \sum_{j \in S} p_{i,j}(a) W(j) \right]$$

Where $W(j)$ is the expected discounted cost on π from time step 1 onwards. Note that $W(j) \geq \alpha V^*(j)$. Thus, we get -

$$\begin{aligned} V_\pi(i) &= \sum_{a \in A} P_\pi(a) \left[c(i, a) + \sum_{j \in S} p_{i,j}(a) W(j) \right] \\ &\geq \sum_{a \in A} P_\pi(a) \left[c(i, a) + \alpha \sum_{j \in S} p_{i,j}(a) V^*(j) \right] \\ &\geq \min_{a \in A} \left[c(i, a) + \alpha \sum_{j \in S} p_{i,j}(a) V^*(j) \right] \end{aligned}$$

This holds for all policies. Thus,

$$V^*(i) = \inf_{\pi \in \Pi} V_\pi(i) = \min_{a \in A} \left[c(i, a) + \alpha \sum_{j \in S} p_{i,j}(a) V^*(j) \right]$$

□

Bellman Operators

Prior to going through this, go through some basic information about Banach's fixed point theorem as illustrated in C. Consider a mapping $F: S \rightarrow \mathbb{R}$ that maps a state to a real value. One can think of the value function as a type of this F function. Given this, we define the **Bellman operator**.

Bellman operator

Considering a function $F: S \rightarrow \mathbb{R}$, we define a map $B: F \rightarrow \mathbb{R}$ as follows.

$$B_\pi(F(i)) \stackrel{\text{def}}{=} c_i(\pi(i)) + \alpha \sum_{j \in S} p_{i,j}(\pi(i)) \cdot F(j)$$

The Bellman operator is defined for a particular policy. However, if the policy is one that minimizes the expected cost over all actions, then the operator becomes the **Bellman optimality operator**.

$$B^*(F(i)) \stackrel{\text{def}}{=} \min_{a \in A} \left[c_i(a) + \alpha \sum_{j \in S} p_{i,j}(a) \cdot F(j) \right]$$

Lemma 5.4. *Both B_π and B^* are contraction map in the Banach space $(\mathbb{R}, \|\cdot\|_\infty)$*

Proof. Let us start with the Bellman operator B_π . We need to show that -

$$\exists L > 0 \quad \text{s.t.} \quad \|B_\pi(F) - B_\pi(G)\|_\infty \leq L \|F - G\|_\infty$$

Let us substitute the definition of B in the equation above.

$$\begin{aligned} \|B_\pi(F) - B_\pi(G)\|_\infty &= \max_{i \in S} \left\{ \sum_{j \in S} p_{i,j}(\pi(i)) [c_{i,j}(\pi(i)) + \alpha F(j)] - \sum_{j \in S} p_{i,j}(\pi(i)) [c_{i,j}(\pi(i)) + \alpha G(j)] \right\} \\ &= \alpha \max_{i \in S} \left\{ \sum_{j \in S} p_{i,j}(\pi(i)) [F(j) - G(j)] \right\} \\ &\leq \alpha \max_{i \in S} \left\{ \sum_{j \in S} F(j) - G(j) \right\} = \alpha \|F - G\|_\infty \end{aligned}$$

Therefore, the Bellman operator is a contraction map with factor α . Now, let us start with the Bellman optimality operator.

$$\|B^*(F) - B^*(G)\|_\infty = \max_{i \in S} \left\{ \min_{a \in A} \sum_{j \in S} p_{i,j}(a) [c_{i,j}(a) + \alpha F(j)] - \min_{a \in A} \sum_{j \in S} p_{i,j}(a) [c_{i,j}(a) + \alpha G(j)] \right\}$$

Since there is a min term, the derivation is not straightforward. Assume that $a' \in A$ minimizes the second term, but doesn't minimize the first term.

Note

We are assuming that the same action doesn't minimize both the terms i.e. for F and G .

$$\begin{aligned} \|B^*(F) - B^*(G)\|_\infty &\leq \max_{i \in S} \left\{ \sum_{j \in S} p_{i,j}(a') [c_{i,j}(a') + \alpha F(j)] - \sum_{j \in S} p_{i,j}(a') [c_{i,j}(a') + \alpha G(j)] \right\} \\ &= \max_{i \in S} \left\{ \alpha \sum_{j \in S} p_{i,j}(a') [F(j) - G(j)] \right\} \\ &\leq \alpha \max_{i \in S} \left\{ \sum_{j \in S} F(j) - G(j) \right\} = \alpha \|F - G\|_\infty \end{aligned}$$

Therefore, the Bellman optimality operator is also a contraction map with contraction factor α . $\square$

Value Iteration

This is a way to find the optimal policy that relies on the fact that the Bellman operator is a contraction map. So, to get the optimal value function, one can iteratively solve the Bellman equations as shown in ??.

Algorithm 15 ValueIteration(M, V_0)

Require: MDP $M = (S, A, P, C, \alpha)$, initial estimate V_0

$t = 0$

while True **do**

for $s \in S$ **do**

$V_{t+1}(s) \leftarrow \min_{a \in A} \sum_{s' \in S} p_{s,s'}(a) [c_{s,s'}(a) + \alpha V_t(s')]$

end for

if $V_{t+1} \approx V_t$ **then**

return V_{t+1}

end if

$t \leftarrow t + 1$

end while

Linear Programming Approach

Linear Programming (LP) solves linear problems under constraints and it turns out, MDP policy optimization can also be formulated as a LP. To do so, first we need to understand the following lemma.

Lemma 5.5. Consider the Bellman optimality operator B^* for the value function defined as -

$$B^*(V(i)) = \min_{a \in A} \left[c_i(a) + \alpha \sum_{j \in S} p_{i,j}(a) \cdot F(j) \right]$$

Then, we have -

$$(B^*(V)) \geq V \quad \implies \quad V^* \geq V$$

Proof. From the contraction nature of the Bellman optimality operator, we have -

$$\begin{aligned} V &\leq (B^*(V)) \leq (B^*)^2(V) \leq \dots \\ \implies V &\leq \lim_{n \rightarrow \infty} (B^*)^n(V) \\ \implies V &\leq V^* \end{aligned}$$

□

As a result of the previous lemma, we can now define a **feasible region** where the optimal policy will reside in and the tightest upper bound on these policies in the optimal policy. Thus, we should get V^* by *maximizing* the value functions in this feasible region. Now, this becomes similar to a LP.

$$\begin{aligned} V^* &= \max \sum_{i \in S} V(i) \\ \text{s.t. } V(i) &\leq (B^*(V))(i) \quad \forall i \in S \end{aligned}$$

However, the problem is not a LP since B^* has a min term which makes it non-linear. We can make it a LP with linear constraints as follows -

$$\begin{aligned} V^* &= \max \sum_{i \in S} V(i) \\ \text{s.t. } V(i) &\leq c_i(a) + \alpha \sum_{j \in S} p_{ij}(a)V(j) \quad \forall i \in S, a \in A \end{aligned}$$

State-Action Value Function ((Q)-function)

For a stationary policy, in addition to the value function, one can also define a state-action value function or the (Q)-function. For a policy π , the Q-function gives the discounted cost if at state i , the system takes action a to begin with and then follows π thereafter.

$$Q_\pi(i, a) = c_i(a) + \alpha \sum_{j \in S} p_{ij}(a)V_\pi(j)$$

When we replace $a = \pi(i)$, then Q-function becomes the value function.

$$Q_\pi(i, \pi(i)) = V_\pi(i)$$

For the optimal policy, we define -

$$Q(i, a) = c(i, a) + \alpha \sum_j p_{ij}(a)V^*(j) \quad \implies \quad V^*(i) = \min_a Q(i, a)$$

Policy Iteration

These are the family of algorithms that update policy (directly) in an iterative fashion till they get the optimal policy. Consider a stationary policy f and another policy f^* defined as -

$$f^*(x) = \begin{cases} \arg \min_a [c_i(a) + \alpha \sum_{j \in S} p_{ij}(a)V_f(j)] & \text{if } x = i \\ f(x) & \text{otherwise} \end{cases}$$

Essentially, the policy f^* is the policy that applies the optimal action at time step 1 for state i and then follows the original policy f for the rest of the time steps.

Lemma 5.6. *Given f^* as defined above, we can say that $V_{f^*} \leq V_f$.*

Proof. We prove this using the Bellman operators.

$$\begin{aligned}
(B_{f^*}(V_f))(i) &= c_i(f^*(i)) + \alpha \sum_{j \in S} p_{ij}(f^*(i)) V_f(j) \\
&\leq c_i(f(i)) + \alpha \sum_{j \in S} p_{ij}(f(i)) V_f(j) \quad (\text{From definition of } f^*) \\
&= (B_f(V_f))(i)
\end{aligned}$$

Since f^* applied the optimal action at time step 1 for state i , the Bellman operator will give lesser value for f^* as compared to Bellman operator for f . Since B_{f^*} is a contraction map, we get -

$$\lim_{n \rightarrow \infty} B_{f^*}^n(V_f) = \boxed{V_{f^*} \leq V_f}$$

□

This is the crux of the policy improvement mechanism - *Improving myopically by taking the action that improves upon a single state's will result in a better policy overall.* This leads to the **Policy Iteration** algorithm as illustrated in 16.

Algorithm 16 PI(M, π_0)

Require: A MDP $M = (S, A, P, C, \alpha)$ and an initial policy π_0
Define Improvable states $IS(\pi_0) = \{s \in S : Q_{\pi_0}(s, \pi_0(s)) < \min_{a \in A} Q_{\pi_0}(s, a)\}$
Define Improvable actions $IA(\pi_0) = \{a \in A : Q_{\pi_0}(s, a) < V_{\pi_0}(s)\}$
while $IS \neq \Phi$ **do**
 for $s \in IS(\pi_0)$ **do**
 Pick an action randomly from $a' \in IA(\pi_0, s)$
 $\pi_0(s) = a'$
 end for
 Re-calculate $IS(\pi_0)$ and $IA(\pi_0, s) \forall s \in S$
end while
return π_0

So at each time step, we define states for which the policy is not taking the optimal action yet (IS) and the actions that are in fact optimal for that state (IA). Then, we change the policy to take an action IA for the states in IS. The whole PI algorithm as mentioned in 16 relies on two facts -

- If the IS set of the policy is empty, then it is the optimal policy.
- If the IS set is non-empty, then the policy is not optimal and there exists an optimal policy for M

These facts are not trivial to understand and these are guaranteed by the **Policy improvement theorem**.

Policy Improvement Theorem

Theorem 5.7. *Consider a MDP $M = (S, A, P, C, \alpha)$ and Π is the set of all possible policies for M . Then,*

- *If $IS(\pi) = \Phi$, then $\pi \in \Pi$ is the optimal policy for M*
- *If $\pi' \in \Pi$ is obtained after a round of PI on π , then $\pi' \prec \pi$*

Proof. Let us assume that $IS(\pi) = \Phi$. Then, we can write -

$$\begin{aligned} IS(\pi) = \Phi &\implies \forall \pi' \in \Pi, V^\pi \preceq B^{\pi'}[V^\pi] \\ &\implies \forall \pi' \in \Pi, V^\pi \preceq B^{\pi'}[V^\pi] \preceq (B^{\pi'})^2[V^\pi] \preceq \dots \preceq \lim_{l \rightarrow \infty} (B^{\pi'})^l[V^\pi] \\ &\implies \forall \pi' \in \Pi, V^\pi \preceq V^{\pi'} \end{aligned}$$

Thus, the policy with an empty set of improvable states has the highest value function out of all the policies. Hence, the policy is **optimal**. Now, we shall prove the other direction.

Consider π to be a sub-optimal policy with a non-empty set of improvable states. Suppose, after one iteration of the PI algorithm, we get a new policy π' . Then, we can write -

$$\begin{aligned} B^{\pi'}[V^\pi] &\preceq V^\pi \\ &\implies \lim_{l \rightarrow \infty} (B^{\pi'})^l[V^\pi] \preceq V^\pi \\ &\implies V^{\pi'} \preceq V^\pi \end{aligned}$$

Thus, the PI algorithm will result in the optimal policy no matter which policy is starts with. $\square$

- Howard's Policy iteration - Every improvable state is changed.
- M.S. Random Policy iteration - Randomly changes the improvable state.
- Simple Policy iteration - Changes the improvable state of one single state.

Examples of practical MDPs

Asset selling problem

Machine replacement

In this case, the state space is

$$S = \{0, 1, \dots, \}$$

which highlights the *degradation state* of a machine. The action space

$$A = 0, 1$$

has two actions – replace the machine (1) or keep it (0).

- Keeping a machine incurs a maintenance cost. $L(i)$ is the maintenance cost at state i .
- Buying a new machine incurs a purchase cost as well. Replacement cost is R .

So, if we write $C(i, 1)$, we denote the cost incurred at degradation state i when we replace the machine. Thus, we get -

$$C(i, 1) = R + L(0)$$

If there is no replacement, then

$$C(i, 0) = L(i)$$

The transition matrix is interesting. Suppose there is no replacement, then we write

$$p_{ij}(0) = P_{ij}$$

However, if there is a replacement, then we can write -

$$p_{ij}(1) = P_{i0} \cdot P_{0j} = P_{0j}$$

Note that we haven't imposed a structure of the transitions — each state can transition into any other state. But we can't go from some state i to any state less than i . So we assume

$$\sum_{j=k}^{\infty} P_{ij}$$

increases in i for all $k \geq 0$. Additionally, we assume that the cost function $L(i)$ is a bounded, increasing sequence.

Claim 5.8. *For an increasing function $h(\cdot)$, we say that $\sum_{j=k}^{\infty} P_{ij}h(j)$ is also increasing in i*

This claim makes sense and is obvious since both P_{ij} and $h(j)$ are increasing sequences.

Lemma 5.9. *The optimal value function $V^*(i)$ is increasing in i*

Proof. The easiest way to prove this is that you apply the Bellman operator and performing value iteration. If the value iteration is increasing and since the operator is a contraction map, the value function will also be increasing.

Given the value function at some state n , we can write the value iteration as -

$$V_{n+1}(i) = \min \left(R + L(0) + \alpha \sum_j P_{0j} V_n(j), L(i) + \alpha \sum_j P_{ij} V_n(j) \right)$$

Assuming an initial estimate of $V_0(i) = 0$, we can write the first iteration as -

$$V_1(i) = \min (R + L(0), L(i))$$

Base Step

We can see that $V_1(i)$ is monotone with respect to i .

Inductive Assumption

We assume that $V_n(i)$ is also monotone in i .

Inductive Hypothesis

Since the claim 5.8 is true, we can say from the update step that V_{n+1} is also monotone in i .

Hence, via induction, we can prove that $V^*(i)$ (obtained when $n \rightarrow \infty$) is also monotone in i . □

Now, we can write the Bellman equation as -

$$V^*(i) = \min \left\{ L(i) + \alpha \sum_j P_{ij} V^*(j), R + L(0) + \sum_j P_{0j} V^*(j) \right\}$$

We can see that in the minimum expression, the first term monotonically increases in i and the second term is a constant independent of i . Thus, the optimal policy is to use the machine up to some point i^* where the first term exceeds the second term. Then, the machine is replaced. This is a sort of **threshold policy** – where the action is changed after a certain threshold cost is met.

$$i^* = \max \left\{ i : L(i) + \alpha \sum_j P_{ij} V^*(j) \leq R + L(0) + \sum_j P_{0j} V^*(j) \right\}$$

Expected Time-Average MDPs

In the discounted-cost MDPs, we made no assumption on the structure of the Markov chains (ergodicity, aperiodicity etc.) and we were guaranteed to have an optimal, stationary policy. But now, we no longer have a discounting factor -

$$\mathcal{M} = (S, A, P, C)$$

The goal now is to **minimize** the expected time-average (ETA) cost -

$$\Phi_{\pi}(i) = \limsup_{n \rightarrow \infty} \frac{1}{n+1} \mathbb{E}_{\pi} \left[\sum_{t=0}^n C(X_t, A_t) \mid X_0 = i \right]$$

Which is the ETA cost under policy $\pi \in \Pi$ when starting from state i . A policy π^* is said to be *optimal* if -

$$\Phi_{\pi^*}(i) = \inf_{\pi} \Phi_{\pi}(i) \quad \forall i \in S$$

Question

Is there a guarantee for π^* to exist?

Let us take an example³ with state space

$$S = \{1, 1', 2, 2', \dots\}$$

We also take action space $A = \{1, 2\}$ and the transition probabilities as -

$$\begin{aligned} p_{i,i+1}(1) &= 1 & ; & & p_{i,i'}(2) &= 1 \\ p_{i',i'}(1) &= 1 & ; & & p_{i',i'}(2) &= 1 \end{aligned}$$

Basically, at a natural state, action 1 takes the system to the next natural state, but action 2 takes the system to the primed version of the state. Once at a prime state, there is no escape and we remain in the same state. The cost incurred is given by -

$$C(i, \cdot) = 1 \quad ; \quad C(i', \cdot) = \frac{1}{i}$$

Now that the MDP is defined, let us look at the working. Consider the case where the policy takes only action 1. Then,

$$\Phi_{\pi}(1) = \limsup_{n \rightarrow \infty} \frac{1}{n+1} \mathbb{E}_{\pi} \left[\sum_{t=0}^n 1 \right] = 1$$

Now in another policy, we have some probability to take action 2. Now, assume that state $\bar{n}$ is the smallest state where we take action 2. Then,

$$\begin{aligned} \Phi_{\pi}(1) &= \limsup_{n \rightarrow \infty} \frac{1}{n+1} \mathbb{E}_{\pi} \left[\bar{n} - 1 + \sum_{t=\bar{n}}^n C(X_t, A_t) \right] \\ &= \limsup_{n \rightarrow \infty} \frac{1}{n+1} \left[\bar{n} - 1 + \frac{p_{\bar{n}}}{\bar{n}} \right] \\ &\geq \frac{p_{\bar{n}}}{\bar{n}} > 0 \end{aligned}$$

As we change $\bar{n}$, the expected cost will change. Since there is no fixed $\bar{n}$, there is no fixed minimum cost and thus we can always come up with another stationary policy that will result in lower ETA cost. Thus,

THERE IS NO OPTIMAL POLICY.

³Since we are taking an example, the answer should clearly be NO since we don't prove using examples.

Question

Is there a guarantee for π^* to exist consider only stationary policies?

Let us take another example⁴ -

$$\begin{aligned} S &= \{1, 2, \dots\} \\ A &= 1, 2 \\ p_{i,i+1}(1) &= p_{i,i}(2) = 1 \\ C(i, i) &= 1 \quad , \quad C(i, 2) = \frac{1}{i} \end{aligned}$$

Now in this case, consider a stationary policy that always plays action 1. Then, the ETA cost will be 1.

On the other hand, if we have a stationary policy that picks the action 2 for the first time in state n , we remain in state n . Since the policy is stationary (at every state, it takes a deterministic action), the policy taking action 2 at n will keep taking action 2 and remaining at state n . Thus, we get the ETA cost as $\frac{1}{n}$.

We can keep choosing higher or higher n and get better policy with lower cost.

In conclusion, for MDPs that have no discounting and aim to minimize ETA, there is no stationary policy that will be optimal. Instead, one can induce some form of non-stationarity (like take action on each state conditionally) to get an optimal policy.

Consider a MDP which has the following specifications -

1. State space S is the set of natural numbers (countably infinite)
2. Action space A is the set $\{1, 2\}$.
3. Transitions T . At state i , action 1 goes to state $i + 1$ with probability 1. On the other hand, action 2 remains at state i with probability 1.
4. The cost C . At state i , action 1 incurs cost 1. However, state action 2 incurs a cost $\frac{1}{i}$.

Now suppose the policy is the following - “At state i , take action 1 with probability $\frac{1}{2^i}$. With the rest of the probability, take action 2”

This policy is a non-stationary optimal policy that incurs an expected average cost as 0

Proof. Consider a Markov Decision Process (MDP) defined by:

- **State Space:** $S = \mathbb{N}$ (The set of natural numbers).
- **Action Space:** $A = \{1, 2\}$.
- **Transitions:**
 - At state i , action 1 $\rightarrow i + 1$ with probability 1.
 - At state i , action 2 $\rightarrow i$ with probability 1.
- **Costs:** $C(i, 1) = 1$ and $C(i, 2) = \frac{1}{i}$.
- **Policy π :** At state i , take action 1 with probability 2^{-i} , and action 2 with probability $1 - 2^{-i}$.

⁴Hint: answer is No again :(

The expected cost incurred in state i under policy π is:

$$\begin{aligned} C_\pi(i) &= P(a = 1|i) \cdot C(i, 1) + P(a = 2|i) \cdot C(i, 2) \\ &= \frac{1}{2^i}(1) + \left(1 - \frac{1}{2^i}\right) \left(\frac{1}{i}\right) \\ &= \frac{1}{2^i} + \frac{1}{i} - \frac{1}{i \cdot 2^i} \end{aligned}$$

The number of time steps T_i spent in state i follows a geometric distribution with success probability $p_i = \pi(a = 1|i) = 2^{-i}$. The expected time spent in state i is:

$$E[T_i] = \frac{1}{p_i} = 2^i \quad (6)$$

The time average cost G is given by the limit of the ratio of total expected cost to total expected time:

$$G = \lim_{n \rightarrow \infty} \frac{\sum_{i=1}^n E[\text{Cost in state } i]}{\sum_{i=1}^n E[T_i]} \quad (7)$$

Let W_n be the total expected cost and S_n be the total expected time up to state n :

- $S_n = \sum_{i=1}^n 2^i = 2(2^n - 1)$
- $W_n = \sum_{i=1}^n E[T_i] C_\pi(i) = \sum_{i=1}^n 2^i \left(\frac{1}{2^i} + \frac{1}{i} - \frac{1}{i \cdot 2^i} \right) = \sum_{i=1}^n \left(1 + \frac{2^n - 1}{i} \right)$

Applying the Stolz–Cesàro theorem:

$$\begin{aligned} G &= \lim_{n \rightarrow \infty} \frac{W_n - W_{n-1}}{S_n - S_{n-1}} \\ &= \lim_{n \rightarrow \infty} \frac{1 + \frac{2^n - 1}{n}}{2^n} \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{2^n} + \frac{2^n - 1}{n \cdot 2^n} \right) = 0 \end{aligned}$$

The time average cost of this MDP under policy π is $\mathbf{0}$. □

It is clear that with some amount of non-stationarity, the policy behaves optimally for the MDP with no discount factor. However, there is a condition under which an optimal stationary policy does indeed exist.

Theorem 5.10. *Given expected average cost for $M(S, A, P, C)$, if there exists a function $h \in \mathcal{B}(S)$ and a scalar g such that -*

$$g + h(i) = \min_{a \in A} \left[c_i(a) + \sum_{j \in S} p_{ij}(a) h(j) \right] \quad \forall i$$

Then, there exists optimal stationary policy π^ such that -*

- $\pi^*(i) = \arg \min_{a \in A} \left[c_i(a) + \sum_{j \in S} p_{ij}(a) h(j) \right]$
- $\Phi_{\pi^*}(i) = g \quad \forall i$

If you look closely, the h is analogous to the value function in the discounted MDPs and the equation is similar to the Bellman equations. If there is a non-discounted MDP for which these values of g, h can be found, then a stationary policy can be found which is optimal and gives 0 ETA cost.

Proof. Let us denote the history $H_t = (X_0, A_0, \dots, X_t, A_t)$. For any policy π , we say that -

$$\mathbb{E}_\pi \left[\sum_{t=1}^n (h(X_t) - \mathbb{E}_\pi \{h(X_t) \mid H_{t-1}\}) \right] = 0$$

This is because

$$\mathbb{E}_\pi \{h(X_t) \mid H_{t-1}\} = \mathbb{E}_\pi \{h(X_t)\}$$

Note

We are already using the fact that we have found a valid g, h for the theorem.

However,

$$\begin{aligned} \mathbb{E}_\pi \{h(X_t) \mid H_{t-1}\} &= \sum_{j \in S} h(j) \cdot \mathbb{P}_{X_{t-1}, j}(A_{t-1}) \\ &= C(X_{t-1}, A_{t-1}) + \sum_{j \in S} h(j) \cdot \mathbb{P}_{X_{t-1}, j}(A_{t-1}) - C(X_{t-1}, A_{t-1}) \\ &\geq \min_{a \in A} \left[C(X_{t-1}, a) + \sum_{j \in S} h(j) \cdot \mathbb{P}_{X_{t-1}, j}(a) \right] - C(X_{t-1}, A_{t-1}) \\ &= g + h(X_{t-1}) - C(X_{t-1}, A_{t-1}) \end{aligned}$$

Substituting this back, we get -

$$\begin{aligned} \mathbb{E}_\pi \left[\sum_{t=1}^n (h(X_t) - g - h(X_{t-1}) + C(X_{t-1}, A_{t-1})) \right] &\geq 0 \\ \mathbb{E}_\pi [h(X_1) - g - h(X_0) + C(X_0, A_0) + h(X_2) - g - h(X_1) + C(X_1, A_1) + \dots] &\geq 0 \\ \mathbb{E}_\pi [h(X_n) - h(X_0) - ng] + \sum_{t=1}^n C(X_{t-1}, A_{t-1}) &\geq 0 \end{aligned}$$

Thus, we get -

$$g \leq \frac{\mathbb{E}_\pi[h(X_n)]}{n} - \frac{\mathbb{E}_\pi[h(X_0)]}{n} + \frac{1}{n} \mathbb{E}_\pi \left[\sum_{t=1}^n C(X_{t-1}, A_{t-1}) \right]$$

Therefore, as $n \rightarrow \infty$, the expected time average cost will be -

$$g \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}_\pi \left[\sum_{t=1}^n C(X_{t-1}, A_{t-1}) \right] = \Phi_\pi$$

□

Hence, having a valid value of g, h ensures that there is a stationary policy that can be optimal. One thing to note (that we also highlighted earlier) is that the expression looks similar to the Bellman equations, but the scaling factor is now being scaled to 1. Recall that for a discounted MDP, we know that -

$$V_\alpha(i) = \min_a \left[c(i, a) + \alpha \sum_{j \in S} p_{ij}(a) V_\alpha(j) \right]$$

So if we go take $\lim_{\alpha \rightarrow 1} V_\alpha(i)$, we expect it to behave the same way as the MDP minimizing ETA cost without discounting. But even though costs are bounded, setting $\alpha = 1$ will cause the costs to blow up asymptotically (summing non-zero values that are not decreasing) and there is no guarantee of convergence for the value function. So can one connect the two MDP paradigms? The answer is a resounding **YES** and we shall show that now. Let us define a value -

$$h_\alpha(i) = V_\alpha(i) - V_\alpha(0)$$

Even though the individual value functions are not bounded, it is highly likely that their differences would be bounded. It is unusual for a policy that reduces the value function for one state and boosts the value function on another state. In other words, we can say that $\lim_{\alpha \rightarrow 1} h_\alpha(i) < \infty$. Now, we can write -

$$h_\alpha(i) + V_\alpha(0) = \min_a \left[c(i, a) + \alpha \sum_{j \in S} \{p_{ij}(a)h_\alpha(j) + V_\alpha(0)\} \right]$$

$$h_\alpha(i) + (1 - \alpha)V_\alpha(0) = \min_a \left[c(i, a) + \alpha \sum_{j \in S} p_{ij}(a)h_\alpha(j) \right]$$

Note

We can see that in the worst case, $V_\alpha(i) = \frac{C}{1-\alpha}$ (sum of infinite GP). Thus, taking $(1 - \alpha)V_\alpha(0)$ should be provably bounded and we can hope that the limit of this quantity exists.

Setting $g = (1 - \alpha)V_\alpha(0)$, we get -

$$h_\alpha(i) + g = \min_a \left[c(i, a) + \alpha \sum_{j \in S} p_{ij}(a)h_\alpha(j) \right]$$

Which was the statement we had used in the previous theorem to show the presence of an optimal stationary policy. Thus, setting h as the difference of value functions between the states and a reference state and assuming that $(1 - \alpha)V_\alpha(0)$ is finite in the limit, we can guarantee the presence of a stationary optimal policy. This is formalized in the theorem below.

Theorem 5.11. *If $\exists N < \infty$ such that $|V_\alpha(i) - V_\alpha(0)| < N$ for all α, i , then -*

- *There exists $g \in \mathbb{R}$ and $h \in \mathcal{B}(S)$ such that Bellman equations hold.*
- *For some sequence $\alpha_n \rightarrow 1$, $h(i) = \lim_{n \rightarrow \infty} h_{\alpha_n}(i)$*
- *$g = \lim_{\alpha \rightarrow 1} (1 - \alpha)V_\alpha(0)$*

However, the question now arises – under what conditions would the difference of value functions i.e. h be bounded? Turns out, one can derive *sufficient conditions* for that to occur, by making use of a concept called the **expected hitting time**. Denoted as $M_{i,0}(\pi_\alpha^*)$, the expected hitting time is the minimum time step (in expectation) for the MDP to reach state 0 when it starts from state i under the optimal policy π^* with discount factor α . The expected hitting time is correlated to the bounded property of h , as illustrated in the theorem below.

Theorem 5.12. *Let us suppose there exists some $N < \infty$ such that the expected hitting time $M_{i,0}(\pi_\alpha^*) < N$ for all i, α . Then*

$$|V_\alpha(i) - V_\alpha(0)| < N \quad \forall \alpha, i$$

This means that if the expected hitting time for state 0 from any other state is finite and bounded, then the value of h will also be bounded.

Proof. Let us assume (WLOG) that all the costs are positive⁵. Let us also define -

$$T = \min \{t > 0 \quad \text{s.t.} \quad X_t = 0\}$$

⁵Since the costs are bounded, we can add finite quantities to the costs and make them positive without breaking anything.

Which is the minimum time it takes to reach state 0 for the first time (i.e. actual hitting time). Then, we can write -

$$V_\alpha(i) = \mathbb{E}_{\pi_\alpha^*} \left[\sum_{t=0}^{\infty} \alpha^t C(X_t, A_t) \mid X_0 = i \right]$$

$$V_\alpha(i) = \mathbb{E}_{\pi_\alpha^*} \left[\sum_{t=0}^{T-1} \alpha^t C(X_t, A_t) \mid X_0 = i \right] + \mathbb{E}_{\pi_\alpha^*} \left[\sum_{t=T}^{\infty} \alpha^t C(X_t, A_t) \mid X_0 = i \right]$$

The first term in the expression above is the expected cost it takes to reach state 0 in time T which is a constant, bounded value. On the other hand, the second term is the expected cost from time step T onwards as the MDP runs asymptotically. Since we know that the expected hitting time is bounded by N , the first term is upper bounded by some CN value. The second term is simply the value function. Thus, we can now write -

$$V_\alpha(i) \leq C\mathbb{E}[T] + V_\alpha(0) \leq CN + V_\alpha(0)$$

Thus, we get -

$$V_\alpha(i) - V_\alpha(0) \leq CN$$

Now, let us switch the sign in the subtraction and get the bounde for that expression. We know that the value function of the MDP starting from state i will be more than the expected cost to reach state 0 by time T . Thus, we write the following expressions -

$$\begin{aligned} V_\alpha(i) &\geq \mathbb{E}_{\pi_\alpha^*} [\alpha^T] V_\alpha(0) \\ V_\alpha(0) - V_\alpha(i) &\leq V_\alpha(0) (1 - \mathbb{E}_{\pi_\alpha^*} [\alpha^T]) \\ &\leq \frac{C}{1 - \alpha} (1 - \alpha^{\mathbb{E}_{\pi_\alpha^*}[T]}) \quad (\text{Using Jensen's inequality}) \\ &= \frac{C}{1 - \alpha} (1 - \alpha^N) \leq CN \end{aligned}$$

Therefore, we can say that

$$V_\alpha(0) - V_\alpha(i) \leq CN$$

Using the two expressions, we can finally conclude that if the expected hitting time is bounded by some N , then we get -

$$|V_\alpha(i) - V_\alpha(0)| \leq CN$$

This makes the value of h bounded and guarantees a stationary optimal policy. $\square$

Corollary 1. *If the state space is finite and any stationary policy induce an irreducible Markov chain, then the hypotheses of the theorems hold. This means that for a MDP with finite states and irreducible (implying that there is a non-zero policy to reach any state from any other state), then the hitting time for any state would be finite.*

6 Reinforcement Learning

The RL problem is effectively the MDP problem where only S, A are known, but P, C are **not known**. This is now a much tougher learning problem where the model needs to **explore** the environment and estimate P, C , in an *effective* manner. There are two main types of RL -

- **Off-policy** - The learning algorithm doesn't dictate the policy during learning. Essentially, given a state, action and cost historical data (without actually interacting with the environment), the learner learns an optimal policy.
- **On-policy** - The learning algorithm is exploring the environment based on some policy and it refines the policy as it gathers more experience.

Effectively, off-policy is a type of batch learning and on-policy is a type of online learning. Based on model types, we again have two types of RL problems -

- **Model-based** - The learning algorithm learns the estimates $\hat{P}, \hat{C}$ and then learns an optimal policy.
- **Model-free** - The learner never explicitly learns P, C , but rather directly learns a policy or value function.

Monte-Carlo approach

This is a naive RL algorithm. Basically, look at the history and using the LoLN, estimate $\hat{P}, \hat{C}$.

$$\hat{p}_{i,j}(a) = \frac{\sum \mathbb{1}\{X_t = i, A_t = a, X_{t+1} = j\}}{\sum \mathbb{1}\{X_t = i, A_t = a\}}$$
$$\hat{c}_{i,j}(a) = \frac{\sum C(X_t, A_t) \cdot \mathbb{1}\{X_t = i, A_t = a, X_{t+1} = j\}}{\sum \mathbb{1}\{X_t = i, A_t = a\}}$$

This is an off-policy learning paradigm since the learner actually doesn't explore the environment but uses history to estimate the parameters. On the other hand, if we go for on-policy algorithms, then an exploration-exploitation tradeoff that creeps in.

Assumptions

- The MDPs have a finite state and action space.
- The MDPs are irreducible (each state is reachable from every other state).

Under these assumptions, we can perform on-policy learning using an ε -optimal policy, at each state, the learner explores w.p. ε and exploits (plays optimal action) w.p. $1 - \varepsilon$. However, we also need to have the exploration probability decay with time and hence a valid choice is $\varepsilon_t = \frac{1}{t}$. Using this decay choice, each state is explored infinitely often almost surely.

Q-learning

This is for an off-policy, model-free RL.

$$\hat{Q}(X_t, A_t) \leftarrow \hat{Q}(X_t, A_t) + a_t \left[C_t + \alpha \min_{a'} \hat{Q}(X_{t+1}, a') - \hat{Q}(X_t, A_t) \right]$$

where a_t is a step-size. We can also have an on-policy Q-learning which is nothing but **SARSA** - $(X_t, A_t, C_t, X_{t+1}, A_{t+1})$. The action is still ϵ -greedy -

$$A_t = \begin{cases} \arg \min_{a'} \hat{Q}(X_t, a') & \text{w.p. } 1 - \epsilon_t \\ \text{Random} & \text{w.p. } \epsilon_t \end{cases}$$

The SARSA update is as follows -

$$\hat{Q}(X_t, A_t) \leftarrow \hat{Q}(X_t, A_t) + a_t \left[C_t + \alpha \hat{Q}(X_{t+1}, A_{t+1}) - \hat{Q}(X_t, A_t) \right]$$

Deep Q-learning

The goal here is to train a NN that will learn the Q-table. How do we train this NN? We do not know what the ideal target values would be, so supervised learning objectives are out of the window. Well, first start by looking at the *trace* we have and collect the tuples – (X_t, A_t, C_t, X_{t+1}) . This way, we have created a mini-batch of size B . With this, we can train using SGD (or any other GD paradigm) to minimize a loss function. However, what is the loss function?

We know that for Q-learning, we shall learn the mapping⁶ –

$$(X_t, A_t) \rightarrow C_t + \alpha \min_{a'} Q_{\theta}(X_{t+1}, a')$$

The loss can be kept simple like a squared loss –

$$\sum_B \left[C_t + \alpha \min_{a'} Q_{\theta}(X_{t+1}, a') - Q_{\theta}(X_t, A_t) \right]^2$$

However, both the target quantity and input are parameterized by θ , which is not easy to optimize. Basically, the input and target are both being adapted at the same time. The hack for this is to perform **two-time scale approximation** –

$$\sum_B \left[C_t + \alpha \min_{a'} Q_{\tilde{\theta}}(X_{t+1}, a') - Q_{\theta}(X_t, A_t) \right]^2$$

Where θ is updating at a faster rate and then after a few iterations, $\tilde{\theta}$ is updated.

TD learning

Temporal difference learning has a goal of learning of the value function for a given stationary policy π .

⁶ θ represents the parameters of the NN.

A Expectation of double summation

Lemma A.1.

$$\mathbb{E} \left[\sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1 - \hat{Y}_{j,t})^2 \right] \leq Kn$$

Proof.

$$\begin{aligned} \mathbb{E} \left[\sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1 - \hat{Y}_{j,t})^2 \right] &= \mathbb{E} \left[\sum_{t=1}^n \sum_{j=1}^K p_{j,t} (1 + \hat{Y}_{j,t}^2 - 2\hat{Y}_{j,t}) \right] \\ &= \sum_{t=1}^n \mathbb{E} \left[1 - 2 \sum_{j=1}^K p_{j,t} \frac{\mathbb{1}_{A_t=j} \cdot Y_t}{p_{j,t}} + \sum_{j=1}^K p_{j,t} \frac{\mathbb{1}_{A_t=j} \cdot Y_t^2}{p_{j,t}} \right] \\ &= \sum_{t=1}^n \mathbb{E} \left[1 - 2Y_t + \sum_{j=1}^K Y_{j,t}^2 \right] \\ &= \sum_{t=1}^n \mathbb{E} \left[(1 - Y_t)^2 + \sum_{j \neq A_t} Y_{j,t}^2 \right] \leq Kn \end{aligned}$$

□

B Probability basics

Convergence of random sequences

Consider a random sequence X_n and a limiting value X . Then,

$$X_n \xrightarrow{\text{a.s.}} X \quad \text{if } \mathbb{P} \left(\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \right) = 1$$

We define convergence in probability (weak convergence) as -

$$X_n \xrightarrow{P} X \quad \text{if } \lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \epsilon) = 0 \quad \forall \epsilon > 0$$

We define convergence in distribution as -

$$X_n \xrightarrow{D} X \quad \text{if } \lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq x) = \mathbb{P}(X \leq x) \quad \forall x$$

Convergence a.s. $\implies$ Convergence in P $\implies$ Convergence in D

Note

For a.s. and weak convergence, we require X_n and X to belong to the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. However, convergence in distribution does not need this assumption to hold.

Limit theorems

Let X_n be a sequence of i.i.d random variables with mean $\mu \in \mathbb{R} < \infty$. Then, we can define a sequence -

$$S_n = \sum_{i=1}^n X_n$$

Then, we can state -

$$\begin{aligned} \text{SLoLN} &\implies \frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu \\ \text{WLoLN} &\implies \frac{S_n}{n} \xrightarrow{P} \mu \end{aligned}$$

Essentially, LoLN states that S_n scales as $S_n \approx n\mu + o(n)$ as n increases. However, what is the behaviour of $o(n)$? This is where **CLT** comes into play. Assume that $\text{Var}(X_i) = \sigma^2 < \infty$ for all i . Then,

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \xrightarrow{D} \mathcal{N}(0, 1)$$

Note

The error of the LoLN approximation scaled by $\sqrt{n}\sigma$ converges to the standard normal distribution. It helps us characterize $o(n)$.

$$\mathbb{P}(\sqrt{na} \leq S_n - n\mu \leq \sqrt{nb}) \xrightarrow{n \rightarrow \infty} \mathbb{P}(Z \in [a, b]) \quad Z \sim \mathcal{N}(0, \sigma^2)$$

$$\mathbb{P}(S_n > na) = \mathbb{P}(S_n - n\mu > n(a - \mu)) = \mathbb{P}\left(\frac{S_n}{n} - \mu > (a - \mu)\right) \xrightarrow{P} 0$$

The above follows from WLoLN.

Concentration inequalities

Markov's inequality

Suppose $X \geq 0$ is a random variable, then -

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a} \quad \forall a > 0$$

For non-negative random variables $X \geq 0$, we get -

$$\int_0^\infty \mathbb{P}(X \geq x) dx = \mathbb{E}[X]$$

Use this to prove Markov's inequality. We can also write -

$$\begin{aligned} \mathbb{P}(S_n > na) &= \mathbb{P}(S_n - n\mu > n(a - \mu)) \\ &\leq \mathbb{P}(|S_n - n\mu| > n(a - \mu)) = \mathbb{P}(|S_n - n\mu|^2 > n^2(a - \mu)^2) \\ &\leq \frac{\mathbb{E}[(S_n - n\mu)^2]}{n^2(a - \mu)^2} = \frac{\text{Var}(S_n - n\mu)}{n^2(a - \mu)^2} \\ &= \frac{n\sigma^2}{n^2(a - \mu)^2} = \frac{\sigma^2}{n(a - \mu)^2} = O\left(\frac{1}{n}\right) \end{aligned}$$

Chebyshev's inequality

Chernoff bound (Tightest bound)

This is the Cramer-Chernoff property discussed in SLT class.

$$\mathbb{P}(Y \geq y) = \mathbb{P}(e^{SY} \geq e^{Sy})$$

Since the exponent is monotone, this transformation is valid. Also, the random variable is now non-negative, so we can apply Markov's inequality.

$$\mathbb{P}(e^{SY} \geq e^{Sy}) \leq \frac{\mathbb{E}[e^{SY}]}{e^{Sy}} = e^{-Sy} \cdot \text{MGF}_S(Y)$$

The above is true for all S . Taking infimum gives us the Chernoff bound.

$$\mathbb{P}(Y \geq y) \leq \inf_{S>0} e^{-Sy} \cdot \text{MGF}_S(Y)$$

This holds only if first exponential moment $\mathbb{E}[e^{SY}] = \text{MGF}_S(Y)$ is finite for some $S > 0$. This is equivalent to saying Y is **light-tailed**.

Note

In the Cramer-Chernoff approach, we use exponential transformation to get non-negative random variable while also preserving the order. We can also perform the following transformation -

$$\mathbb{P}(Y \geq y) = \mathbb{P}(Y^{2S} \geq y^{2S})$$

Where $S \in \mathbb{Z}$. This is essentially Chebyshev's inequality.

Now consider $S_n = \sum_{i=1}^n X_i$. Then, using Chernoff bounds, we get -

$$\mathbb{P}(S_n \geq na) \leq e^{-n \sup_{S>0} [sa - \ln(\text{MGF}_X(S))]}$$

If X is a light-tailed, then $\sup_{S>0} [sa - \ln(\text{MGF}_X(S))] > 0$

Note

Chernoff bound is very specific to the random variable since it relies on the MGF will relies on the fact that we know the distribution of X . However, most times all we know are the bounded regions of random variables, and not the exact distribution. Thus, we need a more generic bounds that are arguable looser. This is where Hoeffding's inequality comes in.

Bounded Distributions

In most cases, we are not aware about the distribution from which we sample the random variables, but we know that X is bounded over $[a, b]$. Then, to get a bound (like Chernoff) in this case (where the MGF is not known), we use **Hoeffding's inequality**.

Hoeffding's inequality

Say $\{X_i\}_{i \geq 1}$ is a sequence of independent⁷ random variables where $a_i \leq X_i \leq b_i$ a.s. for all i . Also define $S_n = \sum_{i=1}^n X_i$. Then, we can write for all $t \geq 0$,

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) \leq \exp\left(\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

⁷need not be identically distributed.

One special case would be if X_i are identically distributed as well with $\mathbb{E}[X_i] = \mu$ and $a \leq X_i \leq b$ a.s. for all i . Then -

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = n\mu$$

Then, we get -

$$\begin{aligned}\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) &= \mathbb{P}\left(\sum_{i=1}^n X_i - n\mu \geq t\right) \\ &= \mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \geq \frac{t}{n}\right) \leq \exp\left(\frac{-2t^2}{n(b-a)^2}\right) \\ &= \mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \geq \epsilon\right) \leq \exp\left(\frac{-2n\epsilon^2}{(b-a)^2}\right)\end{aligned}$$

Proof. We shall be using Hoeffding's lemma as stated in 1.4. Now, using Cramer-Chernoff technique and Markov's inequality, we get -

$$\begin{aligned}\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) &= \mathbb{P}\left(e^{\lambda(S_n - \mathbb{E}[S_n])} \geq e^{\lambda t}\right) \leq \mathbb{E}\left[e^{\lambda(S_n - \mathbb{E}[S_n])}\right] \cdot e^{-\lambda t} \\ &= \left\{\prod_{i=1}^n \mathbb{E}\left[e^{\lambda(X_i - \mathbb{E}[X_i])}\right]\right\} \cdot e^{-\lambda t}\end{aligned}$$

Using 1.4, we get -

$$\mathbb{E}\left[e^{\lambda(X_i - \mathbb{E}[X_i])}\right] \leq e^{\frac{1}{8}(\lambda^2(b-a)^2)}$$

Substituting this and then completing the squares, we can prove Hoeffding's inequality. □

Note

The detailed proof of Hoeffding's lemma and Hoeffding's inequality is given in SLT. Check those notes for more information.

Confidence Interval

Consider the case where we have expectation of n i.i.d random variables p and a ERM estimatory $\hat{p}$. Then, we can get from Hoeffding's inequality that -

$$\mathbb{P}(\|p - \hat{p}\| \geq \epsilon) \leq 2e^{-2n\epsilon^2} = \delta$$

Then, we can write -

$$p \in \left[\hat{p} - \sqrt{\frac{\ln\left(\frac{2}{\delta}\right)}{2n}}, \hat{p} + \sqrt{\frac{\ln\left(\frac{2}{\delta}\right)}{2n}}\right] \quad \text{w.p.} \geq 1 - \delta$$

This is called the **Confidence Interval** with the lower and upper confidence bounds (LCB and UCB).

Unbounded Distributions - Sub-Gaussian random variables

Recall that for a Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$, we have -

$$\text{MGF}_S(X) = \mathbb{E}[e^{SX}] = e^{\left(\mu S + \frac{S^2 \sigma^2}{2}\right)}$$

σ -sub Gaussian

A random variable Y is said to be σ -sub Gaussian if -

$$\mathbb{E}[e^{S(X-\mathbb{E}[X])}] \leq e^{\left(\frac{S^2\sigma^2}{2}\right)} \quad \forall S \in \mathbb{R}$$

We know from 1.4, for any bounded random variable between $[a, b]$ a.s., we have -

$$\mathbb{E}[e^{S(X-\mathbb{E}[X])}] \leq e^{\left(\frac{S^2(b-a)^2}{8}\right)}$$

Thus, we see that all bounded random variables are $\frac{b-a}{2}$ - sub Gaussian.

Note

The class of sub Gaussian random variables includes all the Gaussian random variables as well.

Lemma B.1. Consider X to be a σ -sub Gaussian random variable. Then, for all $t \geq 0$, we get -

- $\mathbb{P}(X - \mathbb{E}[X] \geq t) \leq e^{\left(\frac{-t^2}{2\sigma^2}\right)}$
- $\text{Var}(X) \leq \sigma^2$
- cX would be a $|c|\sigma$ -sub Gaussian.

Proof. TBFL

□

Lemma B.2. Consider $X_1 \sim \sigma_1$ -sub Gaussian and $X_2 \sim \sigma_2$ -sub Gaussian are independent random variables. Then $X_1 + X_2 \sim \sqrt{\sigma_1^2 + \sigma_2^2}$ -sub Gaussian.

Proof. TBFL

□

Theorem B.3. Consider a sequence of i.i.d. random variables $\{X_i\}_{i=1}^n$ that are σ -sub Gaussian. Also, define $S_n = \sum_{i=1}^n X_i$. Then, for all $t \geq 0$, we get -

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) \leq \exp\left(\frac{-t^2}{2n\sigma^2}\right)$$

Proof. TBFL

□

Sub-Exponential random variables

Sub-Exponential RVs

A RV X is called (σ, α) -sub exponential if -

$$\mathbb{E}\left[e^{S(X-\mathbb{E}[X])}\right] \leq e^{\left(\frac{S^2\sigma^2}{2}\right)}$$

for all $|S| < \frac{1}{\alpha}$

Note that we are not restricting the condition over all S like we did for sub-Gaussian.

Note

Sub-Gaussian RV $\subset$ Sub-Exponential.

Lemma B.4. *The following statements are equivalent -*

1. X is sub-exponential
2. $\exists c_0 > 0$ s.t. $\mathbb{E}[e^{SX}] < \infty \forall |S| \leq c_0$
3. $\exists c_1, c_2 > 0$ s.t. $\mathbb{P}(|X| \geq x) \leq c_1 e^{-c_2 x} \forall x \geq 0$

Proof. Taking a suitable value of c_0 , it is trivial to see that -

$$X \text{ is sub-exponential} \implies \exists c_0 > 0 \text{ s.t. } \mathbb{E}[e^{SX}] < \infty \forall |S| \leq c_0$$

Now, we can see from Taylor series expansion that -

$$\mathbb{E} \left[e^{S(X - \mathbb{E}[X])} \right] = 1 + 0 + \frac{\text{Var}(X) \cdot S^2}{2} + o(S^2)$$

We also know from Taylor series expansion that -

$$e^{\left(\frac{S^2 \sigma^2}{2}\right)} = 1 + \frac{S^2 \sigma^2}{2} + o(S^2)$$

Now, we can see that if we choose $\sigma^2 > \text{Var}(X)$, then -

$$\mathbb{E} \left[e^{S(X - \mathbb{E}[X])} \right] \leq e^{\left(\frac{S^2 \sigma^2}{2}\right)}$$

in a neighbourhood around the origin. So, we have proven $1 \iff 2$.

Now, we shall try to show $2 \implies 3$. Let us consider $S \in (0, c_0)$ which guarantees the finiteness of the MGF thanks to claim 2. Then, we get -

$$\mathbb{P}(X \geq x) = \mathbb{P}(e^{SX} \geq e^{Sx}) \leq \mathbb{E}[e^{SX}] \cdot e^{-Sx}$$

This follows from the Chernoff bounds. In a similar sense, we can choose $S \in (-c_0, 0)$ which would give us -

$$\mathbb{P}(X \leq -x) \leq \mathbb{E}[e^{SX}] \cdot e^{-Sx}$$

Thus, we get -

$$\mathbb{P}(|X| \geq x) \leq 2e^{-Sx}$$

Which shows that $2 \implies 3$. Now let us try to prove the other way. Fix some $S > 0$. Then, we can see that for a non-negative random variable e^{SX} , we get -

$$\begin{aligned} \mathbb{E}[e^{SX}] &= \int_0^\infty \mathbb{P}(e^{SX} > t) dt \leq 1 + \int_1^\infty \mathbb{P}\left(X > \frac{\ln t}{S}\right) dt \\ 1 + &= \int_1^\infty c_1 e^{-\frac{c_2}{S} \ln t} dt = 1 + \int_1^\infty c_1 t^{-\frac{c_2}{S}} dt < \infty \quad \forall S < \frac{c_2}{2} \end{aligned}$$

In other words, the MGF is finite and hence we have shown that $3 \implies 2$. DO THE SAME FOR $S < 0$ AND GET AN IDENTICAL BOUND. Hence, we get -

$$1 \iff 2 \iff 3$$

□

Lemma B.5. If X is a (σ, α) sub-exponential distribution, then -

$$\mathbb{P}(X - \mathbb{E}[X] \geq x) \leq \begin{cases} e^{-\frac{x^2}{2\sigma^2}} & \text{For } 0 \leq x \leq \frac{\sigma^2}{\alpha} \\ e^{-\frac{x}{2\alpha}} & \text{For } x > \frac{\sigma^2}{\alpha} \end{cases}$$

Proof. For some $S \in (0, \frac{1}{\alpha})$, we can write -

$$\mathbb{P}(X - \mathbb{E}[X] \geq x) = \mathbb{P}(e^{S(X - \mathbb{E}[X])} \geq e^{Sx}) \leq e^{\frac{S^2\sigma^2}{2}} \cdot e^{-Sx} = e^{\frac{S^2\sigma^2}{2} - Sx}$$

Optimizing this bound, we get two cases.

Case 1 Here, $\frac{1}{\alpha} > \frac{x}{\sigma^2}$ which implies $x < \frac{\sigma^2}{\alpha}$. For this case, the optimal choice of S would be $S^* = \frac{x}{\sigma^2}$. Substituting back, we get -

$$\mathbb{P}(X - \mathbb{E}[X] \geq x) \leq e^{-\frac{x^2}{2\sigma^2}}$$

Case 2 Here, $\frac{1}{\alpha} < \frac{x}{\sigma^2}$ which implies $x > \frac{\sigma^2}{\alpha}$. For this case, the optimal choice of S would be $S^* = \frac{1}{\alpha}$. Substituting back, we get -

$$\mathbb{P}(X - \mathbb{E}[X] \geq x) \leq e^{\frac{\sigma^2}{2\alpha^2} - \frac{x}{\alpha}} \leq e^{-\frac{x}{2\alpha}}$$

These two cases prove the lemma. □

Lemma B.6. Consider $\{X_i\}_{i=1}^n$ to be i.i.d. (σ, α) -sub exponential RVs. Define S_n as the sum of the X_i . Then,

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq x) \leq \begin{cases} e^{-\frac{x^2}{2n\sigma^2}} & \text{For } 0 \leq x \leq \frac{n\sigma^2}{\alpha} \\ e^{-\frac{x}{2\alpha}} & \text{For } x > \frac{n\sigma^2}{\alpha} \end{cases}$$

Proof. TBFL □

Radon-Nikodym Derivative

A tuple of set of events and σ -algebra $(\Omega, \mathcal{F})$ is called a **measurable space**. This means that we can define a measure on this space. If we define a **probability measure** on this space, then it becomes a **probability space** $(\Omega, \mathcal{F}, \mathbb{P})$.

Consider 2 probability measures, $\mathbb{P}$ and $\mathbb{Q}$, defined on the same measurable space $(\Omega, \mathcal{F})$. Then, $\mathbb{P}$ is said to be **absolutely continuous** w.r.t $\mathbb{Q}$ for any $A \in \mathcal{F}$ if -

$$\mathbb{Q}(A) = 0 \implies \mathbb{P}(A) = 0 \iff \mathbb{P}(A) > 0 \implies \mathbb{Q}(A) > 0$$

If $\mathbb{P}$ is absolutely continuous w.r.t $\mathbb{Q}$, then $\exists$ a measurable function f such that for any $A \in \mathcal{F}$, we get -

$$\mathbb{P}(A) = \int_A d\mathbb{P} = \int_A f d\mathbb{Q}$$

This serves as the basis for **Radon-Nikodym derivative**. We can write it in terms of expectation as well -

$$\mathbb{E}_{\mathbb{P}}(1_A) = \mathbb{E}_{\mathbb{Q}}\left(1_A \cdot \frac{d\mathbb{P}}{d\mathbb{Q}}\right)$$

So if we consider some function g that is $\mathbb{P}$ -measurable, then -

$$\int g d\mathbb{P} = \int g \left(\frac{d\mathbb{P}}{d\mathbb{Q}}\right) d\mathbb{Q}$$

Example

Consider a finite probability space where -

$$\Omega = \{x_1, \dots, x_n\} \quad ; \quad \mathcal{F} = \mathcal{P}(\Omega)$$

where we define two measures— $\mathbb{P}$ and $\mathbb{Q}$ —such that -

$$\begin{aligned} \mathbb{P}(\{x_i\}) = p_i \quad \forall i \quad \text{s.t.} \quad p_i > 0 \ \& \ \sum_{i=1}^n p_i = 1 \quad \forall i \\ \mathbb{Q}(\{x_i\}) = q_i \quad \forall i \quad \text{s.t.} \quad q_i > 0 \ \& \ \sum_{i=1}^n q_i = 1 \quad \forall i \end{aligned}$$

If $\mathbb{P}$ is absolutely continuous w.r.t $\mathbb{Q}$, then $p_i = 0 \implies q_i = 0$ for all i . Here, we can get the Radon-Nikodym derivative as follows -

$$\frac{d\mathbb{P}}{d\mathbb{Q}}(x_i) = \frac{p_i}{q_i}$$

KL-Divergence

Consider $\mathbb{P}$ and $\mathbb{Q}$ be two probability measures over the same measurable space with $\mathbb{P}$ being absolutely continuous w.r.t $\mathbb{Q}$. Then,

$$\text{KL}(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{\mathbb{P}} \left[\log \left(\frac{d\mathbb{P}}{d\mathbb{Q}} \right) \right] = \int \log \left(\frac{d\mathbb{P}}{d\mathbb{Q}} \right) d\mathbb{P}$$

Note

$\mathbb{P}$ being absolutely continuous w.r.t $\mathbb{Q}$ doesn't imply that $\mathbb{Q}$ is also absolutely continuous w.r.t $\mathbb{P}$. Thus, both Radon-Nikodym derivative and KL-Divergence are **not symmetric**.

By convention, we say that if $\mathbb{P}$ is **not** absolutely continuous w.r.t $\mathbb{Q}$, then we take the KL-Divergence to be ∞ (even though it is technically not defined).

Example

Consider a finite probability space where -

$$\Omega = \{x_1, \dots, x_n\} \quad ; \quad \mathcal{F} = \mathcal{P}(\Omega)$$

where we define two measures— $\mathbb{P}$ and $\mathbb{Q}$ —such that -

$$\begin{aligned} \mathbb{P}(\{x_i\}) = p_i \quad \forall i \quad \text{s.t.} \quad p_i > 0 \ \& \ \sum_{i=1}^n p_i = 1 \quad \forall i \\ \mathbb{Q}(\{x_i\}) = q_i \quad \forall i \quad \text{s.t.} \quad q_i > 0 \ \& \ \sum_{i=1}^n q_i = 1 \quad \forall i \end{aligned}$$

Then,

$$\text{KL}(\mathbb{P}, \mathbb{Q}) = \sum_{i=1}^n p_i \log \left(\frac{p_i}{q_i} \right)$$

Note

KL-Divergence between two random variables is the same as the KL-Divergence between their probability distributions.

Example

Consider $X \sim \text{Bern}(p)$ and $Y \sim \text{Bern}(q)$. Then,

$$\text{KL}(X, Y) = p \log \left(\frac{p}{q} \right) + (1 - p) \log \left(\frac{1 - p}{1 - q} \right)$$

Example

Consider $X \sim \mathcal{N}(\mu_1, \sigma^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma^2)$. Then, the Radon-Nikodym derivative becomes -

$$\frac{dF_X}{dF_Y}(x) = \frac{\exp \left[-\left(\frac{(x - \mu_1)^2}{2\sigma^2} \right) \right]}{\exp \left[-\left(\frac{(x - \mu_2)^2}{2\sigma^2} \right) \right]} \implies \text{KL}(X, Y) = \frac{(\mu_1 - \mu_2)^2}{2\sigma^2}$$

C Banach Fixed Point Theorem

To illustrate the Banach fixed point theorem, we need to first define something called a contraction mapping.

Contraction Mapping

A mapping $Z : X \rightarrow X$ is called a *contraction map* with a contraction factor of l if $\forall u, v \in X$, we have -

$$\|Z(v) - Z(u)\| \leq l \|v - u\|$$

This is very similar to the definition of Lipschitz continuity of an operator Z . Suppose we can find a point $x^* \in X$ such that -

$$Z(x^*) = x^*$$

i.e. the point remains unchanged after being operated upon by the operator Z . Then, the point x^* is called a **fixed point**.

Banach's Fixed Point Theorem

Theorem C.1. Let $(X, \|\cdot\|)$ be a Banach space^a and let $Z : X \rightarrow X$ be a contraction mapping with the contraction factor of $l \in [0, 1)$. Then,

- Z has a unique fixed point $x^* \in X$
- For all $x \in X$ and $m \geq 0$, we have -

$$\|Z^m(x) - x^*\| \leq l^m \|x - x^*\|$$

^aA Banach space is a complete vector space that has a norm defined on it.

1. Write the proof of 1.3
2. Derive the regret lower-bound in the proof of 1.3
3. Prove 1.4
4. Convince yourself on the lower bound in the regret analysis of 3
5. Fill in the missing steps of the upper bound in the regret analysis of 3
6. Write the substitutions in Regret analysis for 3.
7. Solve Question 1
8. Prove that the “Predicting Real sequence pt. 2” is effectively the same as the “Prediction of real sequences”
9. Check for Formulation 4 why ℓ' is convex with respect to p
10. For formulation 4, verify that $\ell'(X'_t, Y'_{i,t}) = \ell(X_t, Y_{i,t})$
11. Prove Eq. 5
12. Check why loss are independent and why we can apply Hoeffding's inequality
13. Derive the expression for $\text{Var}(\hat{Y}_{j,t} | I_{t-1})$
14. Properly prove the expected regret bound for 5
15. Prove the inequality $1 + x \leq e^x \leq 1 + x + x^2$
16. **First Homework (Due - 23/01/2025)**
17. Prove Markov's inequality in B
18. In Markov's inequality, instead of using second moment, what if we take fourth moment? Will we get a tighter bound?
19. Add Chebyshev's inequality in B
20. Get examples of light & heavy tailed random variables.
21. Prove the Chernoff bound for $S_n = \sum_{i=1}^n X_i$
22. Prove B.1
23. Prove B.2
24. Prove B.3
25. Show that if X is sub-Gaussian, then X^2 need not be sub-Gaussian.
26. Properly write the proof of B.4.
27. Properly write the proof of B.5.
28. Prove B.6
29. Prove 3.1
30. Prove 3.2
31. Check regret bounds for 8 and 9

32. Check how things change if we use $\delta = \frac{1}{t^3}$ in the regret bound of 9.
33. Read up on KL divergence and KL-UCB.
34. Perform regret analysis for 10
35. Prove the BH-inequality

References

- [1] Tor Lattimore and Csaba Szepesvári. *Bandit algorithms*. Cambridge University Press, 2020.